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Timothy M. Lenton

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Land and ocean carbon cycle feedback effects on global warming in a simple Earth system model

By TIMOTHY M. LENTON, *Institute of Terrestrial Ecology, Edinburgh Research Station, Bush Estate, Penicuik, Midlothian EH26 0QB, UK*

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ABSTRACT

A simple Earth system model is developed by coupling a box model of the global carbon cycle to an energy-balance approximation of global temperature. The model includes a range of feedback mechanisms between atmospheric CO₂, surface temperature and land and ocean carbon cycling. It is used to assess their effect on the global change being driven by anthropogenic CO₂ emissions from fossil fuel burning and land-use change. When tuned to reach the 1990 level of atmospheric CO₂, the model CO₂ predictions for 1832–1990 are reasonably close to ice-core and instrumental records, observed global warming of ~0.6 K from 1860–1990 is accurately predicted and the land and ocean carbon sinks for the 1980s are close to IPCC central estimates. The ocean sink is reduced by ~0.3 GtC yr⁻¹ when the ocean surface is assumed to warm at the same rate as global surface temperature. Land and oceanic carbon sinks are predicted to be growing at present and hence buffering the rate of rise of atmospheric CO₂. In the basic model, the current land carbon sink is assumed to be due to CO₂ fertilisation of photosynthesis. The slight warming that has occurred enhances soil respiration (carbon loss) and net primary productivity (carbon uptake) by similar amounts. When the model is forced with a “business as usual” (IS92a) emissions scenario for 1990–2100 followed by a linear decline in emissions to zero at 2200, CO₂ reaches a peak of 985 ppmv in 2170 and temperature peaks at +5.5 K in 2180. Peak CO₂ is ~135 ppmv higher than suggested by IPCC for the same forcing, principally because global warming first suppresses the land carbon sink then generates a land carbon source. When warming exceeds ~4.5 K, soil respiration “overtakes” the CO₂ fertilisation of NPP, triggering a release of ~70 GtC from terrestrial ecosystems over ~100 years. When the effects of temperature on photosynthesis, respiration and soil respiration are removed, peak levels of CO₂ are reduced by ~100 ppmv and peak temperature by ~0.5 K. Distinguishing separate soil carbon pools with different residence times does not significantly alter the timing of the switch to a land carbon source or its effect on peak CO₂, but it causes the source to persist for longer. If forest re-growth or nitrogen deposition are assumed to contribute to the current land carbon sink, this implies a weaker CO₂ fertilisation effect on photosynthesis and generates a larger future carbon source. Peak CO₂ levels are also sensitive by about ±80 ppmv to upper and lower limits on the temperature responses of photosynthesis, plant respiration and soil respiration. By forcing the model with a range of future emission scenarios it is found that the creation of a significant land carbon source requires rapid warming, exceeding ~4.5 K, and its magnitude increases with the rate of forcing. The carbon source is greatest for the most rapid burning of the largest reserve of fossil fuel. It is concluded that carbon loss from terrestrial ecosystems may significantly (~10%) amplify global warming under “business as usual” or more extreme scenarios.

* Corresponding author.
e-mail: tlent@ceh.ac.uk

1. Introduction

Existing predictions of future changes in atmospheric CO₂ and climate may be inaccurate because a chain of different models is often used without considering feedback effects in a consistent way. Carbon cycle “box” models, such as the “Bern” model (Siegenthaler and Joos, 1992), are generally used to predict changes in atmospheric CO₂ for a given scenario of CO₂ emissions. General circulation models (GCMs) are then used to predict climate change based on the predictions of CO₂. Finally, dynamic global vegetation models (DGVMs) and ocean models are used to predict changes in terrestrial and ocean carbon storage from the predicted changes in CO₂ and climate. Lack of closure is implied when changes in natural carbon sinks and sources are predicted that should affect the evolution of atmospheric CO₂, but were not included in the original model used to predict CO₂. In particular, warming induced decreases in terrestrial carbon storage have been predicted in DGVMs (White et al., 1999) that may be expected to amplify the rise of CO₂. Although some existing box models of the carbon cycle (Wigley, 1993) take into account the effect of CO₂ fertilisation on the land carbon sink (negative feedback), they do not deal with warming induced changes in carbon storage on the land (potential positive feedback).

Such issues may be addressed with a new generation of “Earth system models” (ESMs) that are being developed by integrating models of the carbon cycle and the climate. For example, amplification of increasing CO₂ due to warming induced decrease in the uptake of CO₂ by the ocean (Sarmiento et al., 1998) has recently been quantified in a coupled model (Joos et al., 1999). Carbon cycle models are currently being interactively coupled within 3-D general circulation models of the climate (GCMs) to allow comprehensive simulations of future global change (Foley et al., 1998). Intermediate complexity ESMs are also being developed based on, for example, the CLIMBER model (Ganopolski et al., 1998a; Ganopolski et al., 1998b), with the aim of making simulations over glacial–interglacial time scales, and multi-millennial predictions of future global change. A 3rd approach is to develop simple ESMs. These can provide a useful framework to test processes and feedback hypotheses yet to be included in more complex models. They are more tractable to math-

ematical analysis and are potentially applicable to geologic time-scales. Ultimately there will be a suite of models of varying degrees of complexity — a “family” of models in a temporal “nest”.

A range of simple box and box-diffusion models of the ocean–atmosphere carbon cycle (Knox and McElroy, 1984; Oeschger et al., 1975; Sarmiento and Toggweiler, 1984; Siegenthaler, 1983; Siegenthaler and Joos, 1992; Siegenthaler and Wenk, 1984), the terrestrial carbon cycle (Harvey, 1989) and the global carbon cycle (Craig and Holmén, 1995; Jain et al., 1995; Kwon and Schnoor, 1994) have previously been developed. However, climate change has rarely been coupled into such models. One exception is the model of Kwon and Schnoor (1994), which uses a linear relationship between temperature and changes in CO₂, but this ignores the important tendency for the “greenhouse effect” to saturate as CO₂ increases (Houghton et al., 1990). A more complex model of the coupled carbon cycle and climate has been presented by Rotmans and den Elzen (1993), but the functional formulation of CO₂ and temperature effects on terrestrial ecosystem processes are arguably less realistic in that model than in the one developed here. A simpler model has the advantages of being less computationally demanding, exhibiting more transparent behaviour and being more tractable to mathematical analysis.

Here I introduce a simple “Earth system model”, which couples a box model of the carbon cycle to an energy balance representation of global mean surface temperature. The model is designed to be applicable to a wide range of time-scales (10–10⁶ years), but is developed here to study human-induced global change over decades to millennia. Hence, processes that affect atmospheric CO₂ over multi-millennial or longer time-scales, such as CaCO₃ dissolution (Archer et al., 1998) and the weathering of carbonate and silicate rocks (Walker and Kasting, 1992) are not included. CO₂ emissions to the atmosphere from industrial activity and land-use change force the model (and carbon is removed from terrestrial ecosystems due to land-use change). The model is used to assess the effects on atmospheric CO₂ and global temperature of carbon cycle processes and feedback mechanisms reviewed in Section 2. Section 3 then describes the new model, Section 4 explores its response to forcing with historical CO₂ emissions

and Section 5 presents the results of forcing with a range of future emissions scenarios together with a sensitivity analysis.

2. Carbon cycle feedback mechanisms

Feedback occurs when a change in an environmental variable triggers a response that affects the forcing variable. Feedback is “negative” when it tends to damp the initial change and “positive” when it tends to amplify it. The term “feedback” is often used rather loosely in discussions of global change, for example, “positive feedback” is often used to describe any string of processes seen to contribute to change, even when the direct feedback on the forcing variable is weak (or even non-existent). A useful distinction for this study is between direct carbon cycle feedback and indirect CO₂-climate feedback (Schimel et al., 1995).

At present, natural land and ocean carbon sinks are reducing the rate of increase of CO₂ (Schimel et al., 1995). On average, during the 1980s, of the $5.5 \pm 0.5 \text{ GtC yr}^{-1}$ being emitted from fossil fuel combustion and $1.6 \pm 1.0 \text{ GtC yr}^{-1}$ from land-use change, only $3.3 \pm 0.2 \text{ GtC yr}^{-1}$ were accumulating in the atmosphere. The ocean is estimated to have removed $2.0 \pm 0.8 \text{ GtC yr}^{-1}$, and forest re-growth $0.5 \pm 0.5 \text{ GtC yr}^{-1}$. This suggests there was a further $1.3 \pm 1.5 \text{ GtC yr}^{-1}$ “missing” land sink that balanced the carbon budget. Measurements of changes in the O₂/N₂ ratio during 1991–1994 suggest a somewhat stronger land carbon sink of $2.0 \pm 0.9 \text{ GtC yr}^{-1}$ and an ocean sink of $1.7 \pm 0.9 \text{ GtC yr}^{-1}$ (Keeling et al., 1996b). Despite the considerable ranges of uncertainty in such global carbon budgets, most studies agree that significant fractions of anthropogenic emissions are going into both the oceans and the land surface. The questions arise: What processes are responsible? Are they part of feedback mechanisms? How might they change in the future?

2.1. Ocean sink negative feedback on increasing CO₂

The current ocean carbon sink is caused by increasing atmospheric CO₂ and thus represents a negative feedback response. The net transfer of CO₂ between the atmosphere and the surface ocean is driven by the concentration gradient of

CO₂ and depends on its solubility. Broadly speaking, CO₂ out-gasses from the warm low-latitudes and in-gasses in the cold high-latitudes. The net effect of increases in the atmospheric burden of CO₂ is to drive it into the surface ocean. The critical rate-determining step for ocean carbon uptake is the rate of carbon transfer (by advection, diffusion and sinking of biological particles) between the surface and the deep ocean (Siegenthaler and Sarmiento, 1993). Future increases in atmospheric CO₂ will tend to generate a larger ocean carbon sink (Sarmiento et al., 1998). However, the effect will tend to saturate because the ocean buffering capacity decreases as the partial pressure of CO₂ in the ocean increases (Kwon and Schnoor, 1994).

2.2. Ocean sink suppressed by increasing temperature

Warming reduces the solubility of CO₂ and thus tends to reduce its uptake by the ocean. The ocean surface has warmed by $\sim 0.6 \text{ K}$ since 1860 (Houghton et al., 1996), and this warming is predicted to have reduced ocean CO₂ uptake by $\sim 0.2 \text{ GtC yr}^{-1}$ in 1990 (Sarmiento et al., 1998). This effect will increase with future climate warming (Joos et al., 1999; Sarmiento et al., 1998) and will provide a weak positive feedback on the rise of atmospheric CO₂ and global temperature, that detracts from the dominant negative feedback caused by increasing CO₂. Carbon uptake may also be reduced by changes in ocean transport caused by climate change (Sarmiento et al., 1998). Despite uncertainties in these future responses, the ocean carbon sink is probably better understood and constrained than the land carbon sink. Hence greater emphasis is placed in this study on the land carbon sink.

2.3. Processes contributing to the land sink without feedback

There are a number of hypotheses for the cause of the land carbon sink, and the current sink may be due to multiple factors interacting together (Schimel et al., 1995). Forest re-growth and nitrogen fertilisation are two potentially significant contributors that are not part of feedback mechanisms on atmospheric CO₂. Forest re-growth, principally in the Northern Hemisphere, represents

the “recapture” of carbon that was originally released by land-use change (Subsection 4.4). Nitrogen fertilisation of plant growth has been variously estimated to contribute between 0.1 and 2.3 GtC yr⁻¹ (Holland et al., 1997; Peterson and Melillo, 1985; Schindler and Bayley, 1993; Townsend et al., 1996) to the present land carbon sink. However, recent measurements of the fate of nitrogen in temperature ecosystems suggest a maximum effect of 0.25 GtC yr⁻¹ (Nadelhoffer et al., 1999). Nitrogen fertilisation is not part of a feedback on atmospheric CO₂, because CO₂ is not causing the extra growth. However, the ultimate causes of nitrogen fertilisation (human activity) are closely coupled to the causes of increased CO₂ emissions. The effects of carbon sinks due to re-growth or nitrogen deposition on predictions of past CO₂ are discussed in Subsection 4.3 and their effects on future predictions are presented in Subsection 5.2.2.

2.4. Photosynthesis negative feedback on increasing CO₂

The oldest and most widely discussed hypothesis for the main contributor to the land carbon sink is that increases in atmospheric CO₂ fertilise C₃ photosynthesis and carbon uptake (Bacastow and Keeling, 1973). CO₂ and O₂ compete for the enzyme Rubisco, which can act as carboxylase or oxyase. With higher CO₂/O₂ ratio the efficiency of the carboxylase pathway is increased. The resultant enhancement of photosynthetic carbon assimilation is predicted to increase the amount of carbon stored in terrestrial vegetation. This provides a direct negative feedback on the increase in atmospheric CO₂. The effect has traditionally been expressed in terms of a “biota growth factor” (“β” factor), although this approach is less accurate than Michaelis–Menten kinetics for predictive studies (Wigley, 1993). The results of fertilisation experiments on trees (Wullschlegel et al., 1995) and some model studies (Gifford, 1993), suggest that the fertilisation effect is strong enough to generate a current land carbon sink of > 2 GtC yr⁻¹, which is more than sufficient to balance the carbon budget. This would in turn demand a smaller ocean carbon sink than has been predicted, although greater faith is usually placed in ocean than land models. However, other estimates suggest that CO₂ fertilisation is insuffi-

cient to account for the terrestrial carbon sink (Friedlingstein et al., 1995; Randerson et al., 1998).

2.5. NPP enhanced by increasing temperature

There is a rich range of observed responses of ecosystems to temperature variability (Braswell et al., 1997). Increases in the amplitude of the seasonal cycle of CO₂ over the last 30 years plausibly indicate increasing carbon assimilation by vegetation (Keeling et al., 1996a). However, carbon fertilisation alone is unable to explain all of the seasonal effect (Kohlmaier et al., 1989). To account for the discrepancy, a further damping effect on the rise of atmospheric CO₂ has been proposed, whereby increases in temperature stimulate increases in net primary production (NPP) and CO₂ uptake (Braswell et al., 1997; Keeling et al., 1996a) (where, NPP = photosynthesis – plant respiration). Temperature, on a global average, is currently below the optimum for plant growth (Gifford, 1993), and moderate warming has been predicted to increase the rate of photosynthesis more than that of plant respiration, generating an increase in net primary productivity (King et al., 1997). Furthermore, warming tends to lengthen the available growing season, as first indicated by a ~7 day advance in the annual decline of atmospheric CO₂ between the early 1960s and early 1990s (Keeling et al., 1996a). Phenological data for Europe also indicates an advance of 6 days in the start of the growing season and a delay of ~5 days in the end, over the last 30 years (Menzel and Fabian, 1999). Increases in NPP may explain satellite data indicating that negative atmospheric CO₂ anomalies have followed positive temperature and NDVI (normalised difference vegetation index) anomalies by ~2 years, over the last 20 years (Braswell et al., 1997). If warming enhances NPP and CO₂ uptake, tending to decrease atmospheric CO₂, this can provide a weak, indirect negative feedback on temperature.

2.6. Soil respiration enhanced by increasing temperature

Although warming favours plant growth it also enhances heterotrophic respiration, which tends to increase atmospheric CO₂ (Jenkinson et al., 1991; Townsend et al., 1992). The same data that has been used to infer delayed CO₂ uptake after

warming (Braswell et al., 1997) also reveals increased rates of CO₂ growth 1–6 months and 6–8 years after anomalous warm months, that may be caused by enhanced soil respiration (Houghton et al., 1998; Woodwell et al., 1998). Global temperature and CO₂ data for 1958–1992 also show a positive temperature-CO₂ correlation with temperature leading changes in CO₂ by ~6 months (Houghton et al., 1998; Woodwell et al., 1998). Since 1940, positive temperature deviations also appear to have suppressed carbon storage on land with up to a 7-year lag (Houghton, 1995a). **The effect of warming on soil respiration counteracts warming-induced increases in NPP. The net change in ecosystem carbon balance is termed “net ecosystem productivity” (NEP), and defined as $NEP = NPP - \text{soil respiration}$.** Year-to-year switches in the sign of NEP suggest that the effect of warming on soil respiration may already dominate its effect on NPP (Braswell et al., 1997; Houghton et al., 1998; Keeling et al., 1995). In the future this dominance is expected to become more pronounced as (on a global average) soil respiration increases more rapidly with temperature than net primary productivity (Cao and Woodward, 1998; Kwon and Schnoor, 1994). Thus, warming alone can reduce global NEP and cause a net release of CO₂ from the land surface to the atmosphere. This tends to increase atmospheric CO₂ and radiative forcing, potentially providing an indirect positive feedback on global warming.

2.7. Future response of NEP

In the future, the negative effects of temperature on plant and soil respiration will compete with the positive effects of CO₂ fertilisation and temperature on photosynthesis, to modulate NEP. It is not immediately obvious what the result will be. A range of ecosystem, soil and Dynamic Global Vegetation Models (DGVMs) have been used to assess potential changes in terrestrial carbon storage due to increasing CO₂ and climate change. Future global warming has been predicted to cause significant carbon loss from many soil models (Jenkinson et al., 1991; Potter et al., 1993; Schimel et al., 1994; Townsend et al., 1992). However, some of these studies neglect possible changes in carbon input to soils, which can dominate over the effects of temperature on respiration (Saleska et al., 1999). If fluxes of carbon from

vegetation to the soil increase, high-latitude soils have the capacity to act as a carbon sink on decadal timescales (Bird et al., 1996). Old soil organic matter is more tolerant of warming than is sometimes assumed (Liski et al., 1999), but it contributes little to the soil respiration flux or its temperature dependence (Trumbore et al., 1996). Increases in total land carbon storage have been predicted when an ecosystem model was run to equilibrium under changes in CO₂ and climate (King et al., 1997). However, it is the transient, dynamic response of terrestrial ecosystems that is of relevance to future global change (Smith and Shugart, 1993). When vegetation models are forced with increasing CO₂ and climate change they tend to predict an initial growth of the land carbon sink, but as forcing increases the sink “saturates” and then begins to decline in size (Cao and Woodward, 1998), sometimes becoming a carbon source (White et al., 1999). The cause of this behaviour apparently differs amongst models. Saturation of the carbon sink has been attributed to saturation of carbon uptake combined with increasing respiration, especially the temperature effect on soil respiration (Cao and Woodward, 1998). In the hybrid DGVM (Friend et al., 1997) the switch to a carbon source is caused by a decline in living biomass, in particular the death of trees in the Amazon rainforest, caused by climate drying associated with increases in temperature greater than ~4 K after 2050 (White et al., 1999). The future response of NEP is reassessed in the simple model described herein and its effect on atmospheric CO₂ is included explicitly.

3. Model description

The model is intended to be a reasonable first approximation, in which the key components of the carbon cycle and the factors affecting global temperature are represented in a simple but usefully accurate way. A 7-box carbon cycle (Fig. 1) is coupled to an energy-balance approximation of global temperature. Model volumes and areas are given in Table 1 and parameter values are in Table 2.

3.1. Carbon cycle

The initial, pre-industrial steady-state carbon reservoirs (in GtC) and fluxes (in GtC yr⁻¹) of

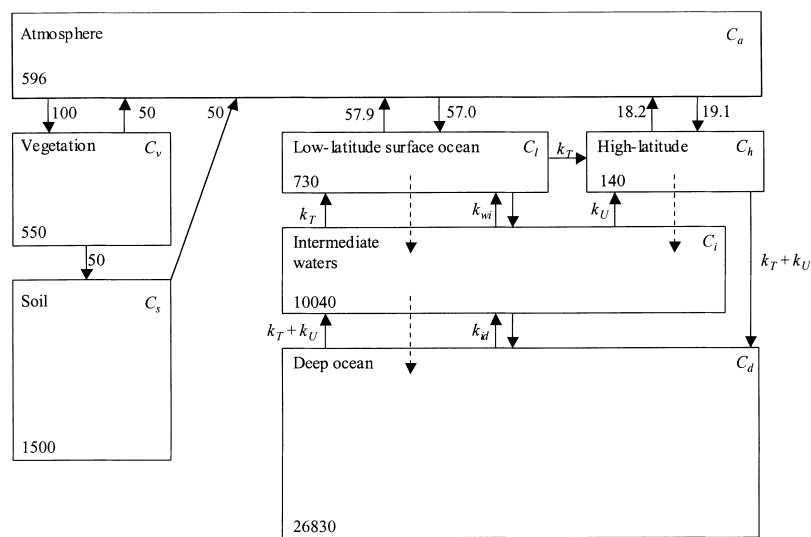


Fig. 1. Model pre-industrial steady-state carbon cycle. Reservoirs are in gigatons of carbon, GtC (10^{15} g), fluxes are in GtC yr⁻¹. The atmospheric reservoir size corresponds to 280 ppmv CO₂. Vegetation and soil reservoir sizes and the fluxes of photosynthesis (P), plant respiration (R_p), turnover (L) and soil respiration (R_s) are from IPCC 1990 (Houghton et al., 1990). The total amount of carbon in the ocean is from Knox and McElroy (1984) and this is apportioned according to the volumes of the boxes and the exchange fluxes. Solid arrows between ocean boxes are water fluxes and dotted arrows are sinking particulate fluxes. Ocean parameter values are given in Table 2. Total carbon exchange between the surface and the deep ocean is ~ 100 GtC yr⁻¹. Air–sea exchange fluxes of CO₂ are those predicted by the model at steady state and compare well with other estimates (Siegenthaler and Sarmiento, 1993).

Table 1. Volumes and areas

Description	Value	Reference
Earth's surface area (a_E)	5.101×10^{14} m ²	(Trenberth, 1992)
volume of the ocean	1.36×10^{18} m ³	(Knox and McElroy, 1984)
area of the ocean	3.6×10^{14} m ²	(Knox and McElroy, 1984)
fractional area of high-latitude surface ocean (A_h)	0.15	(Sarmiento and Toggweiler, 1984)
fractional area of low-latitude surface ocean (A_l)	0.85	(Sarmiento and Toggweiler, 1984)
depth of surface ocean	100 m	(Knox and McElroy, 1984)
mole volume of the atmosphere (k_a)	1.773×10^{20} moles	(Toggweiler and Sarmiento, 1985)
area of vegetated land surface	1.33×10^{14} m ²	(Kwon and Schnoor, 1994)

the model are shown in Fig. 1. In the following equations, reservoirs are expressed in units of moles of C and fluxes in moles of C per unit of time. In the presentation of results, the units are converted back to more familiar measures of each reservoir or flux. The land and ocean components of the carbon cycle can be viewed independently in that they only interact via the atmosphere in the model. In reality there is a flux of

~ 0.5 GtC yr⁻¹ transferred from soils via rivers to the surface ocean, as part of a net cycling of carbon from atmosphere to biota to soils to surface ocean and back to the atmosphere. This net cycling is left out of the model presented here, because it was found to have very little effect on the results, and by ignoring it the consideration of land and ocean carbon sinks is simplified. Terrestrial sedimentation may bury large amounts of organic

Table 2. *Model parameter values*

	Description	Value	Notes
k_T	thermohaline overturning	$6.5 \times 10^7 \text{ m}^3 \text{ s}^{-1}$	tuned to give $\text{CO}_2 = 280 \text{ ppmv}$ at steady state and $\sim 100 \text{ GtC}$ exchange between surface and deep ocean
k_U	high-latitude overturning	$4.87 \times 10^7 \text{ m}^3 \text{ s}^{-1}$	tuned as above
k_{wi}	warm-intermediate exchange	$1.25 \times 10^7 \text{ m}^3 \text{ s}^{-1}$	(Knox and McElroy, 1984)
k_{id}	intermediate-deep exchange	$2.0 \times 10^8 \text{ m}^3 \text{ s}^{-1}$	(Knox and McElroy, 1984)
k_s	piston velocity	1752 m yr^{-1}	(Kwon and Schnoor, 1994)
k_p	photosynthesis rate constant	0.184 yr^{-1}	from estimates of pre-industrial flux and reservoir size
k_r	plant respiration rate constant	0.092 yr^{-1}	from estimates of pre-industrial flux and reservoir size
k_t	turnover rate constant	0.092 yr^{-1}	from estimates of pre-industrial flux and reservoir size
k_{sr}	soil respiration rate constant	0.0337 yr^{-1}	from estimates of pre-industrial flux and reservoir size
k_c	compensation point	29 ppmv	(Gifford, 1993)
K_M	half-saturation point	120 ppmv (145 ppmv)	without (with) ocean warming, tuned to give 1990 atmospheric CO_2
E_a	plant respiration activation energy	$54.83 \text{ kJ mol}^{-1}$	from the Hybrid model (Friend et al., 1997)
k_{MM}	photosynthesis normalising constant	1.478 (1.578)	without (with) ocean warming
k_A	plant respiration normalising constant	8.7039×10^9	
k_B	soil respiration normalising constant	157.072	
S	solar flux	1368 W m^{-2}	(Trenberth, 1992)
A	surface albedo	0.225	(Kasting et al., 1993)
$\tau(\text{CH}_4)$	methane opacity	0.0231	
P_0	water vapour saturation constant	$1.4 \times 10^{11} \text{ Pa}$	(Nakajima et al., 1992)
H	relative humidity	0.6177	tuned to give pre-industrial global average temperature of 288.15 K
c	specific heat capacity	$4.69 \times 10^{23} \text{ J K}^{-1}$	from IS92a forcing of the HadCM3 GCM (Huntingford and Cox, 2000)

All other necessary ocean parameters are as in Knox and McElroy (1984).

carbon (Stallard, 1998) but this only represents a sink of carbon from the atmosphere if it is fuelled by positive NEP, hence NEP is focussed on here.

3.1.1. Land The land carbon cycle model comprises two reservoirs, vegetation (living biomass) (C_v) and soil (including detritus) (C_s). This is the minimal distinction needed to capture the decadal-centennial characteristics of the system (e.g., internal lags). Vegetation is all assumed to use C_3 photosynthesis (the fraction of total biomass due to C_4 and CAM photosynthesis is small (Duursma

and Boisson, 1994)). Vegetation takes up CO_2 from the atmosphere in gross photosynthesis, P and releases it to the atmosphere through “dark” respiration, R_p (comprising both growth and maintenance respiration). Carbon is transferred from vegetation to the soil in the processes of turnover, L (principally litter fall and plant death). Mass balance for the vegetation carbon (biomass) reservoir is given by:

$$\frac{dC_v}{dt} = P(\text{CO}_2, T) - R_p(C_v, T) - L(C_v). \quad (1)$$

The rate constants for photosynthesis (k_p), respiration (k_r) and turnover (k_t) are based on estimates of the pre-industrial fluxes (Fig. 1, Table 2) (Houghton et al., 1990). A wide range of alternative functions for these processes have been explored and it was found that they can often be tuned to be similarly consistent with the historical record. The following approach is intended to be a simple “best guess” at the functional form of the responses, with parameter values that give effects in the mid-range of published estimates. A sensitivity analysis for upper and lower bounds on the temperature responses is presented in Subsection 5.2.1.

Gross photosynthesis is a function of CO_2 and temperature (for $p\text{CO}_{2(a)} > k_c$ and $-15 < \Delta T < 25$, else $P = 0$):

$$P(\text{CO}_2, T) = k_p C_{v(s)} k_{\text{MM}} \frac{p\text{CO}_{2(a)} - k_c}{K_M + p\text{CO}_{2(a)} - k_c} \times \frac{(15 + \Delta T)^2 (25 - \Delta T)}{5625}. \quad (2)$$

$C_{v(s)}$ is the steady-state reservoir size (fixed initially but see Subsection 4.3). ΔT is the deviation of temperature from its initial value of 288.15 K (specified below). $p\text{CO}_{2(a)}$ is the mixing ratio of CO_2 in the atmosphere (in ppmv). Photosynthesis is fertilised by increased atmospheric CO_2 , following Michaelis–Menten kinetics (a hyperbola) but beginning at a CO_2 compensation point, k_c where photosynthetic carbon assimilation goes to zero and no plant growth can occur. K_M is the half-saturation constant for the hyperbola and a measure of the CO_2 fertilisation effect on gross photosynthesis. This parameter is tuned below to give best fit to the historical CO_2 record. k_{MM} simply normalises the carbon fertilisation effect for the initial steady-state value of $p\text{CO}_{2(a)}$. The CO_2 compensation point exists because of the competition of O_2 and CO_2 for the oxygenase or carboxylase activity of the enzyme Rubisco (ribulose biphosphate carboxylase/oxygenase). The carboxylase/oxygenase ratio is similar for different C_3 species (Gifford, 1993). A value of $k_c = 29$ ppmv is used, derived for 15°C from a temperature dependent formula (Gifford, 1993). (The slight increase of the compensation concentration with temperature, does not significantly affect the model results, hence k_c is held constant.) A generalised sigmoidal cubic function is adopted for

the temperature response of photosynthesis (Thornley, 1998). Photosynthesis increases with temperature up to an optimum, usually about 30°C, above which it declines due to increases in photorespiration (the affinity of Rubisco for O_2 increases with temperature more rapidly than its affinity for CO_2) and decreases in electron transport. The temperature function is assumed to go to zero at 0°C and at 40°C, and with a model initial steady-state temperature of 15°C this implies an optimum temperature for photosynthesis of $\sim 27^\circ\text{C}$.

Plant respiration (growth and maintenance) is a function of temperature and biomass:

$$R_p(T, C_v) = k_r C_v k_A e^{-(E_a/RT_s)}. \quad (3)$$

Respiration increases with temperature, T_s (K) following an Arrhenius type response for a general chemical reaction, where E_a is the activation energy and R is the universal gas constant ($8.314 \text{ J mol}^{-1} \text{ K}^{-1}$). This approach and the value for activation energy ($\sim 54.8 \text{ kJ mol}^{-1}$) are taken from the Hybrid model (Friend et al., 1997). k_A normalises the temperature effect. Respiration increases with biomass because greater biomass demands greater maintenance. This dependence buffers changes in the size of the vegetation reservoir that can be forced by changes in CO_2 and temperature.

Net primary production (NPP) is the balance of photosynthesis and respiration ($\text{NPP} = P - R_p$), which at steady state is balanced by turnover. Turnover is assumed to be a constant fraction (k_t) of living biomass:

$$L(C_v) = k_t C_v. \quad (4)$$

This can be thought of as annual litter fall being a constant fraction of standing biomass and a constant fraction of plants dying in a given unit of time. It provides an additional negative feedback on changes in the size of the vegetation carbon reservoir forced by environmental variables (CO_2 and temperature).

Turnover provides input to the soil carbon reservoir. Carbon is lost from the soil reservoir to the atmosphere as CO_2 in soil respiration, R_s . “Soil” respiration includes growth and maintenance respiration of live roots and their associated mycorrhizae as well as respiration of free-living soil heterotrophs in the rhizosphere (the soil immediately surrounding plant roots) and in the rest of

the soil. Mass balance for the soil carbon reservoir is given by:

$$\frac{dC_s}{dt} = L(C_v) - R_s(C_s, T). \quad (5)$$

The rate constant for soil respiration (k_{sr}) is based on an estimate of the pre-industrial flux (Fig. 1 and Table 2). Soil respiration increases with temperature and the size of the soil carbon reservoir (the amount of substrate for free-living heterotrophs):

$$R_s(C_s, T) = k_{sr} C_s k_B e^{-(308.56/(T_s - 227.13))}. \quad (6)$$

The temperature dependence of soil respiration is based on data from a wide range of ecosystem types and temperatures where there was no evidence of soil water limitation (Lloyd and Taylor, 1994). k_B normalises the temperature effect. The functional form is more accurate than traditional “ Q_{10} ” (power law) or Arrhenius type approaches, which Lloyd and Taylor (1994) argued are biased. The dependence of soil respiration on soil mass buffers changes in the size of the soil carbon reservoir that can be forced by changes in turnover or temperature. The effect of dividing the soil into multiple reservoirs with differing turnover times is considered in Subsection 5.2.3.

Net ecosystem productivity (NEP) is defined here as the balance of net primary productivity and soil respiration ($NEP = NPP - R_s = P - R_p - R_s$). In this approach, NEP is a measure of the net change in the combined terrestrial carbon reservoirs and the corresponding exchange of carbon with the atmosphere. When NEP is positive, terrestrial ecosystems are a net carbon sink, when it is negative they are a net carbon source. As the net cycling of carbon via rivers is not included in the model, the sinks and sources presented here should be thought of as the change relative to the pre-industrial state.

3.1.2. Ocean. The key requirement of a carbon cycle model is that it provides reasonable estimates of the land and ocean carbon sinks. In general, 2 or 3 box models of the ocean without diffusion tend to underestimate the current ocean carbon sink, whilst box-diffusion models with outcrop in the high-latitudes tend to over-estimate the ocean carbon sink. Simple box-diffusion and “HILDA” (Siegenthaler and Joos, 1992) schemes can provide estimates of the ocean carbon sink that are in

good agreement with ocean general circulation models, but are more computationally demanding than pure box schemes. Furthermore, values for the diffusion constant inferred from recent CO_2 uptake and used for global change predictions are inconsistent with the low level of atmospheric CO_2 at the last glacial maximum. A three-box ocean model (Sarmiento and Toggweiler, 1984) was found to underestimate current CO_2 uptake (even with appropriately increased carbon exchange between the surface and the deep ocean). A 4-box ocean model including intermediate waters is adopted here because it can provide a reasonable estimate of the ocean carbon sink, without explicitly modelling diffusion, giving the potential for it to be applied to glacial–interglacial and longer time scales.

The ocean carbon cycle model uses the four-box structure of Knox and McElroy (1984), but the areas of the surface boxes and the modelling of air–sea exchange are based on the approach of Sarmiento and Toggweiler (1984). The four ocean “boxes” are the warm, low-latitude surface (w) the cold, high-latitude surface (c), intermediate waters (i) and deep waters (d). Each box contains reservoirs of total carbon dioxide (TCO_2), nitrate (NO_3) and total alkalinity (Talk). Mass balance equations are constructed for the four reservoirs of each of these three tracers. Nitrate and alkalinity are conservative tracers — the total amount of each in the ocean is assumed not to change. The water and particle fluxes connecting the boxes are illustrated in Fig. 1. The water fluxes (Table 2) are increased relative to those suggested by Knox and McElroy (1984) in order to give a total exchange of ~ 100 GtC between the surface and the deep ocean (Houghton et al., 1996; Siegenthaler and Sarmiento, 1993). In reality, some of this carbon flux is due to diffusion, rather than advection. The cold surface box is assumed to cover 15% of the ocean surface area and the warm box, 85% (Sarmiento and Toggweiler, 1984), whereas Knox and McElroy (1984) used 5% and 95%. Other parameter values are as in Knox and McElroy (1984).

Both surface ocean boxes exchange CO_2 with the atmosphere, and the net air–sea exchange flux is of the general form:

$$F_{as} = k_s A_s \rho \beta_s (pCO_{2(a)} - pCO_{2(s)}), \quad (7)$$

where k_s is the piston velocity (0.2 m h^{-1}) and ρ

is the density of seawater (1027 kg m^{-3}). For each surface box, A_s is the area (in m^2), β_s is the Henry's Law constant (solubility in $\text{mol kg}^{-1} \text{ atm}^{-1}$) and $p\text{CO}_{2(s)}$ is the partial pressure of CO_2 (in atm). The flux is defined as positive for net uptake by the ocean. The initial surface box temperatures are 2°C in the high-latitudes ($T_h = 275.15 \text{ K}$) and 22°C in the low-latitudes ($T_l = 295.15 \text{ K}$) (Knox and McElroy, 1984) (these are related to the average global surface temperature below). The cold temperature of the high latitude surface ocean causes net in-gassing of CO_2 (F_{ac} is positive), which can then be transferred to the deep ocean by downward water fluxes (the "solubility pump" of CO_2) and sinking of organic carbon (the "biological pump"). The warmth of the low-latitude surface ocean causes a net out-gassing of CO_2 in the initial state (F_{aw} is negative). The description of seawater carbonate chemistry is updated from that used by Sarmiento and Toggweiler (1984). The various equilibrium constants are calculated as functions of temperature and salinity using the formulae of Goyet and Poisson (1989) and Dickson (1990). The Henry's Law constants (solubilities) of CO_2 in each of the surface boxes (β_w, β_c) are calculated as functions of temperature and salinity. Constant salinity of 34.7‰ is assumed throughout. The partial pressure of CO_2 in each surface box ($p\text{CO}_{2(w)}, p\text{CO}_{2(c)}$) is calculated as a function of carbonate alkalinity and $[\text{TCO}_2]$ (Park, 1969), using the appropriate thermodynamic constants and total borate concentration (a function of salinity). Carbonate alkalinity depends on pH, which is guessed initially and then an iterative procedure is used to find a stable value of pH and a corresponding solution to the carbonate system equations.

3.1.3. Atmosphere. Carbon balance for the atmosphere is given by:

$$\frac{dC_a}{dt} = -P + R_p + R_s - F_{aw} - F_{ac} \quad (8)$$

(recall that F_{aw} is negative in the initial state). The atmospheric mixing ratio of CO_2 , $p\text{CO}_{2(a)}$ is taken as the number of moles of CO_2 in the atmosphere (C_a) divided by the total number of moles of molecules in the atmosphere, k_a (assumed to be

independent of C_a):

$$p\text{CO}_{2(a)} = \frac{C_a}{k_a}. \quad (9)$$

The pre-industrial CO_2 mixing ratio in the atmosphere was $\sim 280 \text{ ppmv}$ (280×10^{-6}) from ice-core data (Etheridge et al., 1998). The model steady-state value is sensitive to the values of ocean circulation parameters and the total amount of carbon in the ocean. Ocean circulation fluxes (Table 2) are adjusted to give precisely 280 ppmv in the steady state and a total carbon transfer from the deep to the surface ocean of $\sim 100 \text{ GtC yr}^{-1}$ (Houghton et al., 1996; Siegenthaler and Sarmiento, 1993).

3.2. Climate: grey-atmosphere approximation

For changes in CO_2 alone, radiative forcing can be approximated by a natural logarithmic function of CO_2 , with a climate sensitivity parameter determining the change in temperature (McGuffie and Henderson-Sellers, 1997). However the "climate" component of the model is intended to be more general and be able to capture changes in global average surface temperature due to changes in albedo, solar flux and the opacity of CO_2 , $\text{H}_2\text{O}_{(v)}$ and CH_4 . For this, a grey atmosphere model is developed (see, for example: Chamberlain, 1980; McKay et al., 1999). The simplifying assumption is made that the atmosphere is transparent to visible radiation and heating only occurs at the surface. Then the net downward flux of radiation (F_d) absorbed at the planet's surface is given by:

$$F_d = \frac{(1-A)S}{4} \left(1 + \frac{3}{4}\tau \right), \quad (10)$$

where A is the surface albedo, S is the incoming solar flux at the top of the atmosphere (1368 W m^{-2}) and τ is the (equivalent grey) vertical opacity of the greenhouse atmosphere. The opacity of each gas is assumed to be independent of the others:

$$\tau = \tau(\text{CO}_2) + \tau(\text{H}_2\text{O}) + \tau(\text{CH}_4). \quad (11)$$

The opacity of CO_2 is assumed to be a function of its mixing ratio. It is derived from the results of varying CO_2 in a radiative-convective climate model (Kasting et al., 1993) where the temperature and water vapour profiles of the atmosphere were

fixed and hence $\tau(\text{H}_2\text{O})$ was constant:

$$\tau(\text{CO}_2) = 1.73(\text{CO}_2)^{0.263}. \quad (12)$$

Methane is kept constant at its pre-industrial level of 650 ppbv (Chappellaz et al., 1990) in the simulations presented here. This gives a constant opacity (Table 2) from a function derived from the same radiative-convective model (L. L. Brown and J. F. Kasting, personal communication).

The opacity of water vapour is assumed to be a function of water vapour pressure, which in turn can be expressed as a function of temperature (the Clausius–Clapeyron equation) and relative humidity:

$$P_{\text{H}_2\text{O}} = H(P_0 e^{-(L/RT)}), \quad (13)$$

where R is the molar gas constant ($8.314 \text{ J mol}^{-1} \text{ K}^{-1}$), L is the latent heat per mole of water ($43\,655 \text{ J mol}^{-1}$), P_0 is a constant for the water vapour saturation curve (Table 2), T is temperature and H is relative humidity. As a first-order approximation, we use the surface temperature and a relative humidity of ~ 0.62 , which is somewhat less than the surface average of ~ 0.77 but gives a water vapour pressure more representative of the whole atmosphere.

$\tau(\text{H}_2\text{O})$ is derived from eq. (13) and the results of a radiative-convective model, with constant $\tau(\text{CO}_2)$, no $\tau(\text{CH}_4)$ and varying solar flux (S), in which feedback on surface temperature is due to variations in $\tau(\text{H}_2\text{O})$ (Fig. 1c of Kasting et al., 1993). For the temperature range $0\text{--}40^\circ\text{C}$ the following is a good approximation to the results of Kasting et al. (1993):

$$\tau(\text{H}_2\text{O}) = 0.0126(P_{\text{H}_2\text{O}})^{0.503}. \quad (14)$$

The surface albedo is kept constant at $A \sim 0.225$, the value required to reproduce the Earth's mean surface temperature of 288.15 K (15°C), for $S = 1368 \text{ W m}^{-2}$, in the radiative-convective model (Kasting et al., 1993). This rather high surface albedo parameterises the effect of clouds. The grey-atmosphere approach then gives a good approximation of the surface temperature predictions of the full radiative-convective model (Kasting et al., 1993).

The globally averaged surface temperature, T_s (K) is calculated in differential form from the energy balance of incoming and outgoing radi-

ation, and the heat capacity of the system:

$$c \frac{dT_s}{dt} = (F_d - \sigma T_s^4) a_E, \quad (15)$$

where σ is the Stefan–Boltzman constant ($5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$), a_E is the Earth's surface area ($5.101 \times 10^{14} \text{ m}^2$) and c is the specific heat capacity, due mostly to the oceans. A value of $c = 4.69 \times 10^{23} \text{ J K}^{-1}$ is used which has been derived from IS92a forcing of the HadCM3 GCM (Huntingford and Cox, 2000). This effective heat capacity is large and varies considerably for different forcing of the GCM. This limitation should be borne in mind in the results where different emissions scenarios are used to force the model. Some heat capacity is needed to produce an expected lag between radiative forcing and global warming on sub-century time-scales. For application to longer time-scales the steady-state temperature solution to (15) can be used. In setting up the initial, pre-industrial steady state of the model, relative humidity is tuned to give $T_s = 288.15 \text{ K}$.

Both incoming and outgoing radiation at the surface depend on surface temperature (T_s) providing two direct geophysical feedbacks. The former effect is due to the dependence of water vapour pressure on temperature, which provides a positive feedback on Earth's surface temperature (Dickinson, 1986; Lashof, 1989). The latter effect is due to the 4th power dependence of long-wave emission on surface temperature, which provides a negative feedback.

The option to relate the temperatures of the high-latitude and low-latitude ocean surfaces directly to the global average surface temperature is included in the model:

$$T_h = T_s - 13, \quad T_l = T_s + 7. \quad (16)$$

The basis for this approach is that during the period 1860–1986, global average sea surface temperatures have risen by approximately the same amount as global mean air temperature ($\sim 0.6 \text{ K}$). These formulae assume that a constant temperature gradient is maintained between the high and the low latitudes and do not deal explicitly with the accumulation of heat in the surface ocean, so they should be viewed as a crude first approximation. On closer examination of the spatial pattern of sea-surface temperature change, large areas of the ocean, including the high latitudes of

the North Atlantic, have actually cooled in recent decades (Houghton et al., 1996). In most of the simulations the surface ocean boxes are kept at their initial temperatures (i.e. eq. (16) is not used), but the effect of ocean surface warming, which reduces the solubility of CO_2 , is considered in Subsections 4.5 and 5.1.

The differential equations are solved with a 4th order Runge–Kutta technique using a time step of the order of 1 year. The time step has to be reduced to 1/4 year in the case of high CO_2 levels achieved in global change experiments but is still much longer than the 1 day used in an existing “simple global carbon model” (Kwon and Schnoor, 1994). This increase in computational efficiency is achieved by not explicitly including the reservoirs of carbon in the marine biota, which turn over very rapidly.

4. Historical forcing, calibration and results

4.1. Historical CO_2 emissions

The model is forced with total CO_2 emissions from industrial activity (fossil fuel burning plus cement production) and land-use change (mostly deforestation). The forcing functions for the period 1800–1990 (Table 3) are based on estimates of industrial, land-use and total emissions from the Carbon Dioxide Information Analysis Center (<http://cdiac.esd.ornl.gov/>). Total CO_2 emissions for 1850–1980 were summed from estimates of industrial emissions (Marland et al., 1998) and estimates of land-use change emissions (Houghton and Hackler, 1998) (Fig. 2). There is a clear inflection in the rate of increase of emissions shortly after the Second World War, and a 2-stage linear fit for the periods 1850–1947 and 1947–1980

provides a good approximation to the overall pattern (Fig. 2). 2-stage linear fits to land-use and industrial emissions are also reasonable (Fig. 2). Exponential envelopes, e.g., Hasselmann et al. (1997), do a poorer job of approximating the estimated emissions pattern. The choice of 1947 should not be taken in a precise sense; the acceleration of CO_2 emissions appears to have occurred over the post war period 1945–1950. Over the period 1800–1850, industrial emissions are estimated to have risen from $\sim 0 \text{ GtC yr}^{-1}$ to $\sim 0.05 \text{ GtC yr}^{-1}$ (Marland et al., 1998), but changes in land-use emissions are uncertain. The estimated land-use flux at 1850 ($\sim 0.35 \text{ GtC yr}^{-1}$) is adopted for 1800–1850, on the basis that land-use emissions were significant prior to 1800 (when the model runs begin), so adopting a smaller value would underestimate their pre-industrial impact. Over the period 1880–1990, estimated fossil fuel emissions rose from $\sim 5.2 \text{ GtC yr}^{-1}$ to $\sim 6.0 \text{ GtC yr}^{-1}$ (Marland et al., 1998), but changes in land-use emissions are uncertain. Total emissions are estimated at $\sim 6.7 \text{ GtC yr}^{-1}$ in 1980 and are assumed to reach 7.4 GtC yr^{-1} in 1990, which is the value at the start of future emissions scenarios (Leggett et al., 1992). This demands a decline in land-use emissions from 1.5 GtC yr^{-1} to 1.4 GtC yr^{-1} from 1980 to 1990.

These simplified emissions scenarios (Table 3) give cumulative total emissions for 1850–1986 of 301 GtC, comprising 198 GtC from industrial activity and 103 GtC from land-use change. These values compare well with IPCC estimates of $312 \pm 40 \text{ GtC}$ total, $195 \pm 20 \text{ GtC}$ industrial, and $117 \pm 35 \text{ GtC}$ land-use (Houghton et al., 1990). Cumulative land-use emissions for 1850–1980 are 94 GtC, which compares well with a separate estimate of 100 GtC (Houghton, 1995b).

4.2. Atmospheric CO_2 record

Model predictions of (annual average) atmospheric CO_2 are compared with two overlapping historical records; 1832–1978 from air trapped in ice-cores at Law Dome, Antarctica (Etheridge et al., 1998) and 1959–1997 from direct atmospheric measurements at Mauna Loa, Hawaii (Keeling and Whorf, 1998) (Fig. 3). The ice-core record is derived from a spline fit (20 year cut-off) of the Law Dome DE08 and DE08-2 ice cores, with an uncertainty of $\pm 1.2 \text{ ppmv}$. The

Table 3. *Estimated historical CO_2 emissions (GtC yr^{-1})*

Year	Fossil fuel	Land-use	Total
1800	0	0.35	0.35
1850	0.05	0.35	0.4
1947	1.2	0.8	2.0
1980	5.2	1.5	6.7
1990	6.0	1.4	7.4

Linear changes in emissions with time are assumed between these values.

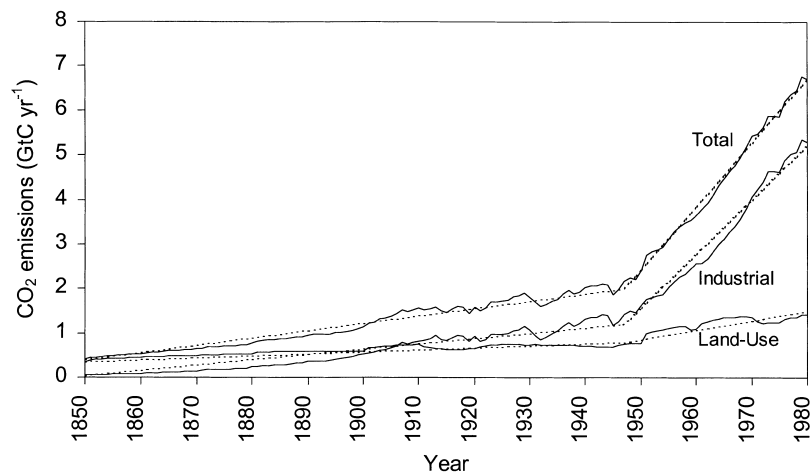


Fig. 2. Estimated industrial, land-use and total carbon dioxide emissions 1850–1980 compared to linear fits. Emissions are in gigatons of carbon per year, GtC yr^{-1} ($10^{15} \text{ g yr}^{-1}$). The solid lines are industrial emissions estimates (from Marland et al., 1998), land-use emissions estimates (from Houghton and Hackler, 1998) and total emissions from summing the two components. The dashed lines are linear fits to these estimates (Table 3), which are extended to cover 1800–1850 and 1980–1990 (as described in the text) and are used to force the model.

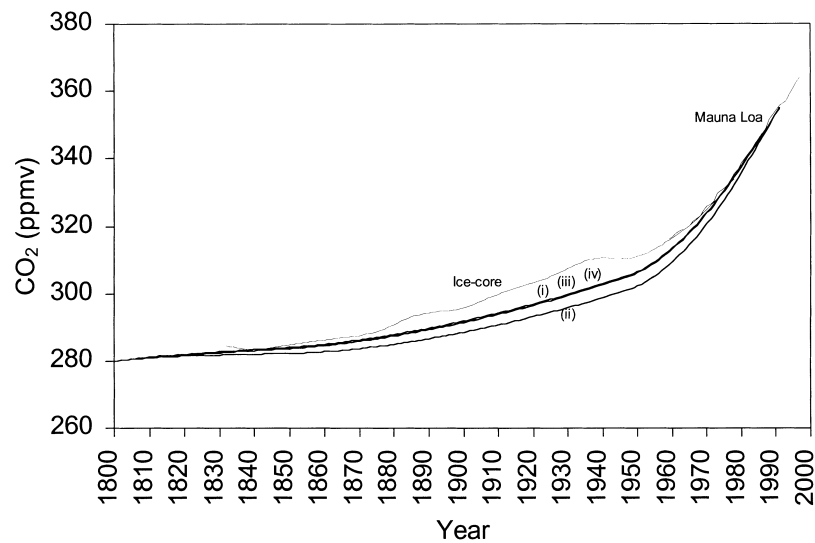


Fig. 3. Atmospheric carbon dioxide records (1832–1997) compared with model predictions (1800–1990). The faint lines are two overlapping historical records; 1832–1978 from air trapped in ice-cores at Law Dome, Antarctica (Etheridge et al., 1998) and 1959–1997 from direct atmospheric measurements at Mauna Loa, Hawaii (Keeling and Whorf, 1998). The bold lines are the predictions of the following versions of the model, in which the CO_2 fertilisation parameter, K_M , or the parameter adjusting steady state biomass, k_D , are tuned to achieve 353 ppmv CO_2 in 1990: (i) No warming of the surface ocean, land-use carbon not removed from the terrestrial reservoirs, $K_M = 120$ ppmv. (ii) Land-use carbon removed from the vegetation reservoir, $K_M = 18$ ppmv and $k_D = 0$. (iii) Land-use carbon removed from vegetation, $K_M = 120$ ppmv and $k_D = 0.27$. (iv) With warming of the surface ocean, $K_M = 145$ ppmv, otherwise as (iii). Predictions of (i), (iii) and (iv) are very similar, but (ii) gives a noticeably poorer prediction.

Mauna Loa record is slightly above the global average, which was 353 ppmv in 1990 (Houghton et al., 1996). Only the historical record up to 1990 is relevant for model calibration. The most important aspects of the historical record to capture for making future predictions are the correct growth rate of CO_2 at the end of the historical forcing (1960–1990) and the correct mixing ratio at 1990. The model is tuned to achieve the latter.

4.3. Tuning of CO_2 response

For comparative purposes, the model is first calibrated when forced with total emissions but without the land-use carbon removed from the terrestrial carbon reservoirs. Although this approach is inconsistent, it has been adopted in other studies, e.g., Craig and Holmén (1995). The temperature responses of photosynthesis, respiration and soil respiration are fixed (as described above) and tuning is restricted to the poorly constrained carbon fertilisation parameter, K_M . In developing the model, many different temperature responses and parameter values were experimented with, but variations in these do not generally improve the fit to the historical CO_2 record. The small historical change in temperature generally produces small increases in net primary production and soil respiration that tend to cancel one another.

Deviations of model CO_2 predictions from the atmospheric record could be due to deficiencies of the model or they could indicate errors in the estimates of CO_2 emissions used to force the model. A value of $K_M = 120$ ppmv (with no warming of the ocean surface) gives a reasonable fit to the record (Fig. 3(i)), although there is a general tendency for the model to under-predict CO_2 in the period 1850–1960 with a maximum deficit of -7 ppmv (c. 1937). This may be because the emissions input to the model are too small during this period. The deficit was found to be unaffected by changes in the relative sizes of the land and ocean carbon sinks. One possible explanation is that the model is over-estimating the carbon fertilisation effect in order to compensate for a missing mechanism that tends to increase the land carbon sink only in the later period (~ 1930 –1990). Estimates of the carbon sink due to deposition of fossil fuel nitrogen over the period 1850–1990 (Townsend et al., 1996) are a roughly constant fraction of

emissions. Hence, including such a sink (as in Subsection 5.2.2) does not improve the fit to the historical CO_2 record. If anything, the effect of nitrogen fertilisation has been saturating in recent decades (Townsend et al., 1996), which would lead to a worse fit. The ice-core record appears to conflict with the emissions estimates used, because it shows a slight decline in atmospheric CO_2 during the Second World War. To achieve this in the model would require a preceding decline in emissions (perhaps related to the Great Depression).

Model predictions of changes in temperature are inevitably much smoother than the historical record (Houghton et al., 1996). However, when a change in CO_2 close to the historical record is achieved in the model, the overall magnitude of warming predicted from 1860–1990 is ~ 0.6 K, in good agreement with observations (Houghton et al., 1996).

4.4. Treatment of land-use change

For consistency, the flux of CO_2 due to land-use change must be removed from the terrestrial carbon reservoirs, and the model behaviour is sensitive to this removal. The flux is mostly deforestation, and hence it is removed from the vegetation carbon reservoir (whether the flux is removed from vegetation only or vegetation and soil makes little difference):

$$\frac{dC_v}{dt} = P(T, \text{CO}_2)C_{v(s)} - R(T, C_v) - T(C_v) - D(t), \quad (17)$$

where $D(t)$ is the land-use change flux. Removing a deforestation flux from the vegetation reservoir has the effect of increasing NEP, because plant respiration, turnover and hence soil respiration depend on the size of the biomass reservoir, which is being depleted by deforestation, whilst the input flux of carbon is independent of the reservoir size. This NEP response can be thought of as forest re-growth, which may account for part of the current carbon sink on land. However, the size of the effect in the model ($\sim 1 \text{ GtC yr}^{-1}$ during the 1980s) is large relative to estimates ($0.5 \pm 0.5 \text{ GtC yr}^{-1}$ (Schimel et al., 1996)) and the response is rather rapid. The historical land-use flux may have been over-estimated. Furthermore, regardless of the tuning of vegetation parameters,

its removal from the land carbon reservoir tends to detract from the fit of model predicted atmospheric CO_2 to historical data (Fig. 3(ii)), as has been noted in a previous study (Craig and Holmén, 1995). In particular, because land-use emissions are a declining fraction of total emissions, their effect on NEP tends to reduce the early growth rate of atmospheric CO_2 more than it reduces the later growth rate. Then if a weaker carbon fertilisation effect is adopted, the increase of CO_2 in the later period (~ 1950 onwards) is too rapid.

The conversion of forest to agricultural land causes a decrease in biomass that persists as long as agriculture does. In regions of the tropics, where land-use involves shifting cultivation this decrease in biomass is short-lived and re-growth can soon begin. However, in other regions conversion to agriculture is more permanent. At a global scale the growth of human population and agricultural land has reduced the biomass of vegetation on the land surface (Vitousek et al., 1997). To represent this in the model, there must be some decrease in $C_{v(s)}$ (steady state biomass) due to deforestation as long as agriculture persists in regions that were once forested:

$$C_{v(s)} = C_{v(0)} - k_D \int_0^t D(t) dt, \quad (18)$$

where $C_{v(0)}$ is the pre-industrial reservoir size (550 GtC), t is time since the start of the model run (1800), and the parameter k_D determines the fraction of biomass loss due to land-use change that is permanent. This approach has the (desired) effect of reducing the tendency for deforestation to enhance NEP. k_D can be tuned and a value of 0.27 returns a good fit to the historical record (Fig. 3(iii)) by effectively factoring out the tendency of land-use change to enhance NEP.

4.5. Predicted carbon sinks

With the above alterations, and without warming of the ocean surface, the model predicts a land carbon sink (NEP) of 1.98 GtC yr^{-1} and an ocean carbon sink of 1.98 GtC yr^{-1} in 1990. Average values for the sinks in the 1980s are 1.89 GtC yr^{-1} on land and 1.82 GtC yr^{-1} in the ocean. These values are close to the central IPCC estimates (Schimel et al., 1996). If the ocean surface is allowed to warm, following eq. (16), $K_M = 145 \text{ ppmv}$ gives a similar fit to the historical record

(Fig. 3(iv)). Then in 1990 the ocean sink is 1.68 GtC yr^{-1} (a decrease of 0.3 GtC yr^{-1}) and the land sink 2.29 GtC yr^{-1} . Corresponding average 1980s sinks are 2.17 GtC yr^{-1} on land and 1.55 GtC yr^{-1} in the ocean. This land estimate is somewhat larger and the ocean estimate correspondingly smaller than those from other (“off-line”) model studies, and measurement based estimates. The ocean sink is suppressed in the model by a lowering of the solubility of CO_2 due to warming. However, this effect has rarely been included in previous model estimates of the ocean carbon sink. A separate, recent estimate is that it decreases the ocean sink by 0.2 GtC yr^{-1} in 1990 (Sarmiento et al., 1998). The model without ocean warming is used as the baseline in the following simulations because it appears to give a more accurate estimate of the current carbon sinks.

5. Future global change

For 1990–2100, a selection of IPCC “emissions scenarios” (Leggett et al., 1992) can be used to force the model (Table 4), but beyond 2100 there are few published suggestions (apart from the model-derived stabilisation scenarios of IPCC 1995 (Houghton et al., 1996)). This is unfortunate, because the inherent lags in the Earth system mean that even if emissions decline dramatically after the year 2100, CO_2 and temperature will continue to rise for some time.

Table 4. *Total CO_2 emissions scenarios for 1990–2100 and land-use component (GtC yr^{-1})*

Year	IS92a	IS92c	IS92e	1990 levels	Land-use
1990	7.4	7.4	7.4	7.4	1.4
1995	7.9	7.2	8.2	7.4	1.4
2000	8.4	7.5	9.1	7.4	1.4
2005	9.2	7.8	10.2	7.4	1.36
2010	9.9	8.1	11.4	7.4	1.32
2015	10.6	8.3	12.6	7.4	1.28
2020	11.4	8.5	13.7	7.4	1.24
2025	12.2	8.8	15.1	7.4	1.2
2050	14.5	7.5	20.1	7.4	0.9
2075	16.3	5.6	27.0	7.4	0.45
2100	20.3	4.6	35.8	7.4	0

Linear changes in emissions with time are assumed between these values.

5.1. Extended "business as usual"

The only suggestion from IPCC for emissions beyond 2100 is extension of the "business as usual" IS92a scenario with a linear decline in emissions from 2100 reaching zero at 2200 (page 324 of IPCC 1995 (Houghton et al., 1996)). Although this is somewhat arbitrary, it is used as the first forcing of the model (scenario 1, Table 5), so that the results can be compared with those mentioned in IPCC 1995. It also serves to illustrate the aforementioned lags in the system. The land-use component is based on that for IS92a, but increased by 0.1 GtC yr^{-1} throughout (and the fossil fuel component correspondingly decreased), so that land-use emissions start at 1.4 GtC yr^{-1} in 1990 and end at 0 GtC yr^{-1} in 2100 (Table 4).

The bold line in Fig. 4 shows the results for the model with no change in ocean surface temperatures ($K_M = 120 \text{ ppmv}$). The predicted temperature change for a doubling of CO_2 from pre-industrial levels (to 560 ppmv) is $\sim 2.5 \text{ K}$ in the dynamic simulation, whilst if the model is allowed to reach equilibrium at doubled CO_2 it is $\sim 2.8 \text{ K}$. Predicted warming due to a doubling of CO_2 from the 1990 level (353 ppmv to 706 ppmv) is $\sim 2.8 \text{ K}$ in the dynamic simulation, whilst the difference in the equilibrium temperature at these CO_2 levels is $\sim 3 \text{ K}$. These values are in the mid-range of GCM estimates (Houghton et al., 1996). CO_2 reaches a peak level of $\sim 985 \text{ ppmv}$ in 2170, which is $\sim 135 \text{ ppmv}$ higher than the $\sim 850 \text{ ppmv}$ in 2160 suggested by IPCC 1995 (although that study also included changes in emissions of CH_4 , N_2O , halocarbons and SO_2). Warming peaks at $+5.5 \text{ K}$ in 2180, which seems plausible given that CO_2 exceeds $3.5 \times$ its pre-industrial level. Peak warming is significantly higher than the $+3.1 \text{ K}$ suggested in IPCC 1995 but compares reasonably with other studies. For example, quadrupling of CO_2 over 140 years has been predicted to cause a

5 K global warming (Manabe and Stouffer, 1993), and the land surface is expected to warm significantly more than the global average (Houghton et al., 1996). The lag of peak temperature behind peak CO_2 is much shorter in the simple model than is generally the case in GCM studies, despite adopting a large value for heat capacity. Two factors tend to reduce the lag. Firstly, incoming radiation in the energy balance equation has a positive feedback dependence on temperature (water vapour feedback) that will tend to accelerate warming. Secondly, outgoing radiation in the energy balance equation has a negative feedback dependence on temperature that tends to take the temperature rapidly toward steady state. By year 3000, CO_2 is stabilising at $\sim 580 \text{ ppmv}$ and temperature at $+3 \text{ K}$. The final "airborne fraction" of CO_2 (the fraction of total CO_2 emissions remaining in the atmosphere) is 22%, which compares well with other models (Siegenthaler and Joos, 1992) (F. Joos, personal communication).

The particularly high peak levels of CO_2 are amplified first by suppression of the land carbon sink and then by generation of a land carbon source. NEP is predicted to peak at $\sim 2.5 \text{ GtC yr}^{-1}$ in 2030 and then decline, becoming a carbon source by 2120. NEP becomes a maximum carbon source of $\sim 1.1 \text{ GtC yr}^{-1}$ in 2150, shortly before CO_2 peaks, and returns to being a weak carbon sink by 2220. Whilst the land is a carbon source it loses a total of 69 GtC . During the initial period when the land is a carbon sink (1800–2110), 329 GtC are gained due to positive NEP, but 221 GtC are lost due to land-use change, a cumulative change of $+108 \text{ GtC}$. Thus, over 60% of the carbon gained by terrestrial ecosystems by 2110 is lost in the next 100 years, before they start to gain carbon again. The peak size of the land carbon sink and the approximate size of the land carbon source are similar to those

Table 5. Long-term composite scenarios

Scenario	1800–1990	1990–2100	Beyond 2100
1	Table 3	IS92a	linear decline to zero at 2200
2	Table 3	IS92c	linear decline to zero at 2200
3	Table 3	IS92e	linear decline to zero at 2200
4	Table 3	IS92a	linear decline using up 4000 GtC
5	Table 3	1990 levels	linear decline using up 4000 GtC
6	Table 3	IS92e	linear decline using up 4000 GtC

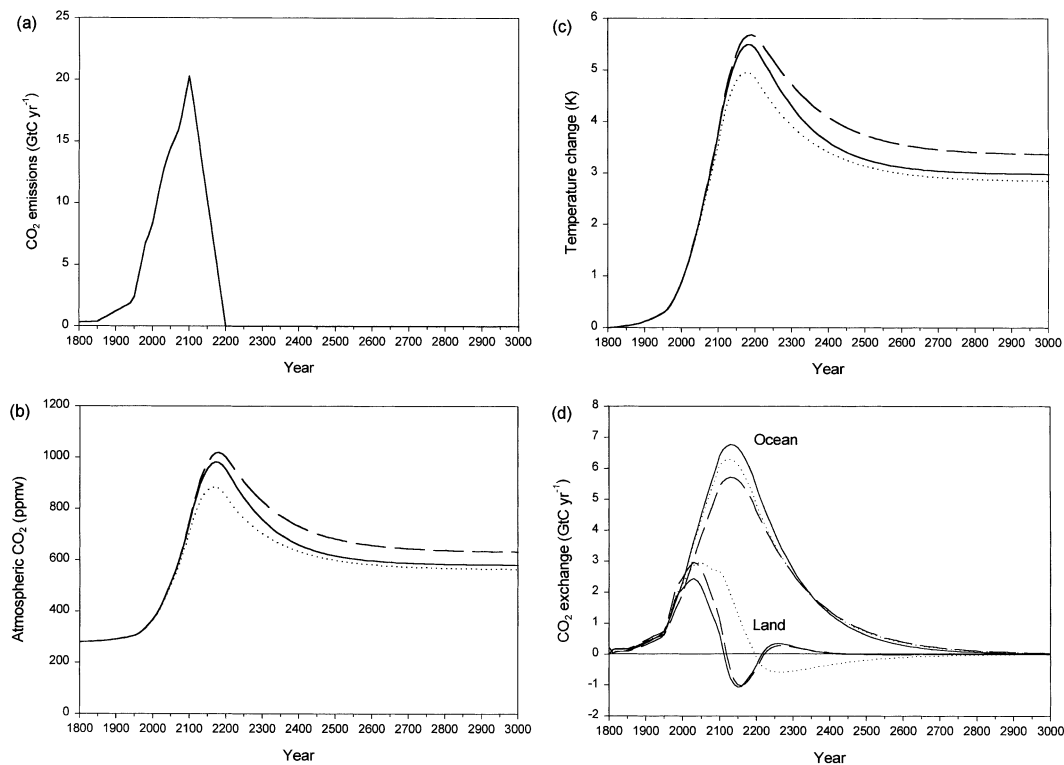


Fig. 4. Effect of extended "business as usual" (IS92a) emissions. Three versions of the model are forced with an extended "business as usual" emissions scenario 1 (detailed in Table 5), as suggested in IPCC 1995 (page 324 of Houghton et al., 1996). In (b)–(d), the solid line is for the model without ocean surface warming (model version (iii) in Fig. 3, $K_M = 120$ ppmv). The dashed line is for the model with ocean surface warming (model version (iv) in Fig. 3, $K_M = 145$ ppmv). The dotted line is for a version of the model (v) identical to (iii) except that the effects of temperature on photosynthesis, plant respiration and soil respiration are removed from equations (2), (3) and (6). (a) Total CO_2 emissions scenario 1. (b) Atmospheric CO_2 . (c) Temperature change relative to pre-industrial level of 288.15 K (15°C). (d) Net CO_2 exchange between the ocean and atmosphere and the land and atmosphere. A positive value represents a carbon sink whilst a negative value represents a carbon source. "Ocean" exchange = $F_{aw} + F_{ac}$. "Land" exchange = $NEP = P - R_p - R_s$.

predicted when the Hybrid DGVM is forced with a CO_2 scenario corresponding to IS92a and climate change predictions from the HadCM2 GCM (White et al., 1999). However, the switch to a carbon source occurs later in the simple model.

The "switch" of NEP from land carbon sink to source is principally due to a changing balance of the effect of CO_2 fertilisation on carbon uptake and the effect of temperature on carbon release from soil respiration. Initially, net carbon uptake occurs as increasing CO_2 fertilises growth, providing direct negative feedback on CO_2 . Net primary production (NPP), living biomass and soil mass

all initially increase, but as temperature increases there is a counteracting increase in soil respiration. Once the rate of increase of soil respiration exceeds the rate of increase of NPP then the land carbon sink starts to shrink. The carbon fertilisation effect on photosynthesis saturates at high CO_2 , tending to slow the increase of NPP (and allow the dependencies of plant and soil respiration on the sizes of the vegetation and soil reservoirs to catch up). Driven by warming, soil respiration eventually "overtakes" net primary productivity causing terrestrial ecosystems to become a net source of carbon to the atmosphere. The "critical threshold" where terrestrial ecosystems become a carbon

source is at $\sim 19.5^\circ\text{C}$ ($+4.5\text{ K}$) corresponding to $\sim 860\text{ ppmv CO}_2$.

The ocean carbon sink more closely tracks the atmospheric burden of CO_2 , becoming the dominant carbon sink after 1990 and continuing to grow to a peak of $\sim 6.8\text{ GtC yr}^{-1}$ in 2130, ~ 30 years after emissions have begun to decline but ~ 40 years before CO_2 peaks. The ocean sink then shrinks but the ocean never becomes a net carbon source. The ocean sink predicted in 2060 is 4.9 GtC yr^{-1} , very similar to that predicted in the “constant biota” control run of a GCM without ocean warming (Sarmiento et al., 1998).

5.1.1. Quantifying positive feedback from ocean warming. The dashed line in Fig. 4 shows the results of the model with ocean surface temperatures changing in proportion to atmospheric temperature and the carbon fertilisation effect correspondingly increased ($K_M = 145\text{ ppmv}$). CO_2 and temperature reach higher peak levels of 1020 ppmv and 5.7 K respectively, and the peaks occur ~ 10 years later. The peak ocean carbon sink is reduced by over 1 GtC yr^{-1} and the predicted value in 2060 of 4.1 GtC yr^{-1} is very similar to that in the “global warming constant-biota” run of the aforementioned GCM (Sarmiento et al., 1998). The residual levels of CO_2 and temperature are amplified more than the peak levels, CO_2 is $\sim 630\text{ ppmv}$ at year 3000 and temperature is $+3.4\text{ K}$. These results indicate that warming-induced reduction in ocean carbon uptake and corresponding increases in atmospheric CO_2 could generate a significant positive feedback on temperature. The results also suggest that the ocean carbon sink exerts the dominant control on residual levels of CO_2 and temperature, but the land carbon sink/source has the greater effect on peak levels of CO_2 and temperature.

5.1.2. Quantifying positive feedback from land warming. The dotted line in Fig. 4 shows the results for a version of the model without ocean warming ($K_M = 120\text{ ppmv}$) in which the effects of temperature changes on photosynthesis, plant respiration and soil respiration are removed, from eqs. (2), (3), and (6). Plants and soil therefore experience a constant temperature, but changing CO_2 . Contrasting the results with those of the baseline model (solid line in Fig. 4) gives an estimate of the importance of warming-induced changes

in NEP for changes in atmospheric CO_2 and temperature. During the historical period (1800–1990), removing the warming effects causes NEP to be slightly increased and predicted CO_2 at 1990 to be reduced by $\sim 1\text{ ppmv}$. Hence, the current land carbon sink in the model is due mostly to CO_2 fertilisation of photosynthesis. The effects of temperature on NPP and soil respiration approximately cancel one another, although there is a slight overall suppression of NEP predicted due to the present warming.

Removing the effects of temperature on NEP reduces future peak levels of CO_2 by $\sim 100\text{ ppmv}$ and peak temperature by $\sim 0.5\text{ K}$ (Table 6). Hence, most of the discrepancy between the peak CO_2 predictions of the baseline model and those in IPCC 1995 is due to including the deleterious effects of warming on NEP. The positive feedback effect due to warming induced suppression of the land carbon sink and creation of a land carbon source amplifies peak global warming by $\sim 11\%$ under this forcing. It has a smaller effect on residual levels of CO_2 and temperature, which are stabilising at $\sim 565\text{ ppmv}$ and $+2.85\text{ K}$ in the run without warming effects. There is only a small residual effect because the net uptake of carbon by terrestrial ecosystems is predicted to be a transient phenomenon, even when they are only forced by increasing CO_2 (dotted line in Fig. 4d). The land becomes a weak source of carbon by 2200, because CO_2 is declining and hence photosynthesis is declining and the land is gradually “relaxing” toward lower carbon storage. Land carbon loss must balance land carbon uptake in a steady state, and because of the faster turnover of carbon in the land system, this balance is achieved more rapidly than for the ocean.

5.2. Sensitivity analysis

We now consider the sensitivity of our predictions for extended “business as usual” (scenario 1 forcing) to aspects of the land carbon cycle model. In each case where the model is altered it is recalibrated by tuning K_M (CO_2 fertilisation) to achieve the 1990 level of atmospheric CO_2 (353 ppmv).

5.2.1. Temperature responses. The temperature responses of photosynthesis (Berry and Björkman, 1980; Mohr and Schopfer, 1995), plant respiration

Table 6. *Amplification of peak CO₂ and temperature due to effects of warming on terrestrial ecosystems*

Scenario	Peak CO ₂ without warming (ppmv)	Peak CO ₂ with warming (ppmv)	CO ₂ amplification due to warming	Peak ΔT without warming (K)	Peak ΔT with warming (K)	ΔT amplification due to warming
1	885	984	11%	4.96	5.50	11%
2	484	500	3%	2.16	2.31	7%
3	1375	1600	16%	7.33	8.20	12%
4	1136	1299	14%	6.29	7.02	12%
5	859	927	8%	4.83	5.22	8%
6	1350	1565	16%	7.22	8.07	12%

(Woodwell, 1983) and soil respiration (Holland et al., 1995; Raich and Schlesinger, 1992) vary considerably amongst plant and soil types and sites, making any global average response subject to uncertainty. The sensitivity of the model predictions to each temperature response was explored by contrasting upper and lower limits with the “best guess” included in the original model. For simplicity the upper and lower limits use a Q_{10} function, which expresses the relative change in rate for a 10°C increase from the reference temperature of 15°C (288.15 K):

$$f(T) = Q_{10}^{(T-15)/10}. \quad (19)$$

Table 7 summarises the upper and lower limit Q_{10} values for each process obtained from a literature search, and the corresponding values of K_M .

For comparison, the temperature responses of plant respiration and soil respiration used in the original model can be reasonably approximated by “best guess” Q_{10} functions over the limited temperature range we are considering (Johnson and Thornley, 1985; Lloyd and Taylor, 1994). The assumed functional response of photosynthesis to temperature has a quite different shape, but a low Q_{10} with an enhanced CO₂ fertilisation effect reproduces a similar NEP response to the original model, allowing some comparison with the upper and lower limit Q_{10} values.

Table 8 summarises the results of the sensitivity analysis. The peak CO₂ prediction of the original model is roughly in the middle of the range of uncertainty for each process. Model predicted peak CO₂ varies by about ± 80 ppmv for extremes

Table 7. *Limits for the temperature responses of terrestrial ecosystem carbon fluxes*

Process	Response	Q_{10}	K_M (ppmv)	Source/notes	Ref.
photosynthesis	upper limit	2.66	108	$E_a = 70 \text{ kJ mol CO}_2^{-1}$ for light saturated photosynthesis	(Mohr and Schopfer, 1995)
	“best guess”	1.5	217	approximates the NEP response of the original model	—
	lower limit	1.1	307	data for C ₃ plants grown in cool conditions	(Berry and Björkman, 1980)
plant respiration	upper limit	3.0	146	value for tropics	(Woodwell, 1983)
	“best guess”	2.15	120	approximation of original formula	—
	lower limit	1.4	87	data for pea seedlings	(Woodwell, 1983)
soil respiration	upper limit	3.8	183	fast cycling fraction of soil	(Trumbore et al., 1996)
	“best guess”	2.4	120	median of reported values, approximation of original formula	(Lloyd and Taylor, 1994; Raich and Schlesinger, 1992)
	lower limit	1.3	69	lowest reported value	(Raich and Schlesinger, 1992)

Table 8. *Sensitivity analysis for extended “business as usual” scenario 1 forcing*

Alteration	Effect	Maximum land carbon sink (GtC yr ⁻¹) (year)	Maximum land carbon source (GtC yr ⁻¹) (year)	Peak CO ₂ (ppmv) (year)
control model	none	2.44 (2030)	−1.07 (2150)	984 (2170)
photosynthesis	upper limit	2.91 (2050)	−0.69 (2230)	889 (2170)
temperature response	lower limit	2.14 (2020)	−2.30 (2140)	1042 (2180)
plant respiration	upper limit	2.25 (2020)	−2.39 (2140)	1038 (2180)
temperature response	lower limit	2.73 (2040)	−0.27 (2210)	926 (2170)
soil respiration	upper limit	2.16 (2020)	−2.95 (2140)	1062 (2180)
temperature response	lower limit	2.81 (2040)	−0.22 (2340)	921 (2170)
re-growth	maximum	1.48 (1990)	−2.47 (2130)	1078 (2170)
	intermediate	1.88 (2010)	−1.79 (2140)	1034 (2170)
N fertilisation	strong	2.35 (2030)	−1.57 (2150)	1003 (2170)
	saturated (1990–2100)	2.14 (2020)	−1.36 (2150)	1005 (2170)
3-box soil	—	1.90 (2020)	−1.13 (2160)	1004 (2170)
3-box soil	respiration of resistant soil independent of temperature	1.93 (2020)	−0.62 (2160)	988 (2170)

of the temperature responses (corresponding to a change in peak temperature of ± 0.4 K). The magnitude of sensitivity is similar for each process, although the response of photosynthesis to temperature is beneficial whilst those of plant respiration and soil respiration are detrimental. The upper limit temperature response of photosynthesis reduces peak CO₂ by nearly 100 ppmv, but this maximum response is only achieved by C₃ plants at low temperatures under light-saturated conditions (Mohr and Schopfer, 1995).

5.2.2. Re-growth and nitrogen fertilisation contributions to the land carbon sink. The current land carbon sink may include contributions from forest re-growth and/or nitrogen fertilisation, which would demand a weaker CO₂ fertilisation effect to balance the carbon budget. The effect of re-growth was mimicked by reducing k_D (although this has a deleterious effect on the fit to the historical CO₂ record). There is a strong trade-off between the contributions of re-growth and CO₂ fertilisation to the land carbon sink. A maximum re-growth sink, corresponding to ~ 1 GtC yr⁻¹ in 1990, is achieved in the model by setting $k_D = 0$ with $K_M = 18$ ppmv (Fig. 3(ii)). An intermediate

sink of ~ 0.5 GtC yr⁻¹ in 1990 is obtained for $k_D = 0.135$ (half its standard value) with $K_M = 61$ ppmv. The total land carbon sink in 1990 is 1.48 GtC yr⁻¹ in the first case and 1.76 GtC yr⁻¹ in the second. The current sink is reduced because with re-growth it is larger in the past.

If re-growth is a significant contributor to the current carbon sink this implies a larger carbon source in the future that will significantly increase peak levels of CO₂ (Table 8). With maximum re-growth, the land carbon sink declines after 1990 becoming a maximum source of -2.47 GtC yr⁻¹ in 2130 that increases peak CO₂ by ~ 95 ppmv. With intermediate re-growth, the land carbon sink peaks at 1.88 GtC yr⁻¹ in 2020 and becomes a maximum source of -1.79 GtC yr⁻¹ in 2140, increasing peak CO₂ by ~ 50 ppmv. However, the current sink due to re-growth is now thought to be at the lower end of the 0.5 ± 0.5 GtC yr⁻¹ range, suggested by IPCC (Lloyd, 1999; Schimel et al., 1995).

Estimates of the carbon sink generated by nitrogen deposition from fossil fuel emissions (Townsend et al., 1996) have a very similar temporal pattern to the fossil fuel emissions themselves (Marland et al., 1998). In the model run of

Townsend et al. (1996) corresponding to the upper end of their best estimate range, the predicted sink is roughly one-eighth of the fossil fuel emissions throughout the period 1850–1990. This does not include any additional carbon sink due to nitrogen deposition from ammonia emissions (Holland et al., 1997) but is still large compared to some other estimates (Nadelhoffer et al., 1999). In a 1st experiment, a strong carbon sink due to nitrogen deposition is introduced that remains a constant one-eighth of fossil fuel emissions. This carbon is simply removed from the atmosphere and added to the vegetation reservoir. The CO_2 fertilisation effect must be reduced accordingly, with $K_M = 104$ ppmv. The total land carbon sink in 1990 is 1.95 GtC yr^{-1} , 0.75 GtC yr^{-1} due to nitrogen fertilisation and 1.2 GtC yr^{-1} due to CO_2 fertilisation. This ignores the likelihood that increasing nitrogen deposition is alleviating nitrogen limitation of ecosystems and hence the corresponding carbon sink is saturating (Townsend et al., 1996). Hence, a second experiment was conducted in which the carbon sink due to nitrogen deposition was held constant from 1990 to 2100 at 0.75 GtC yr^{-1} then reduced linearly to zero at 2200 (with declining fossil fuel emissions).

Including a nitrogen fertilisation contribution to the land carbon sink has little effect on the model predictions, because it is counteracted by a weaker negative feedback due to CO_2 fertilisation (Table 8). In the first experiment the carbon sink due to nitrogen deposition peaks at 2.54 GtC yr^{-1} in 2100, but this is counteracted by a $-2.09 \text{ GtC yr}^{-1}$ carbon source due to the balance of CO_2 fertilisation and temperature effects. Overall NEP reaches a minimum of $-1.57 \text{ GtC yr}^{-1}$ in 2150 and peak CO_2 is increased by ~ 20 ppmv. In the second experiment, NEP reaches a minimum of $-1.36 \text{ GtC yr}^{-1}$ in 2150 and peak CO_2 is increased by a similar amount. Nitrogen fertilisation has less of an effect than forest re-growth because it continues to generate a sink throughout the period of fossil fuel emissions, whereas the sink due to re-growth is assumed to cease with land-use change in 2100 (Table 4).

5.2.3. Subdividing the soil reservoir. The rapidity of predicted changes in land carbon storage may be affected by the use of a single soil reservoir with a relatively short mean residence time of

30 years. In reality, mobile and resistant soil organic carbon reservoirs relax more slowly to equilibrium than this, whilst detritus (litter) responds more rapidly (Schimel et al., 1994). Hence, a version of the model distinguishing fast-turnover detritus, intermediate-turnover mobile soil and slow-turnover resistant soil was developed, based on model 8 of Harvey (1989), and subjected to identical forcing. Fig. 5 shows the initial reservoir sizes and fluxes, which give residence times of 1.8 yr for detritus, 80 yr for mobile soil and 577 years for resistant soil. The turnover flux from eq. (4) was divided into separate input fluxes to each soil box. The transfer fluxes between soil reservoirs were each assumed to be linearly dependent on the donor reservoir size. Initially the respiration fluxes from detritus, mobile soil and resistant soil were each assumed

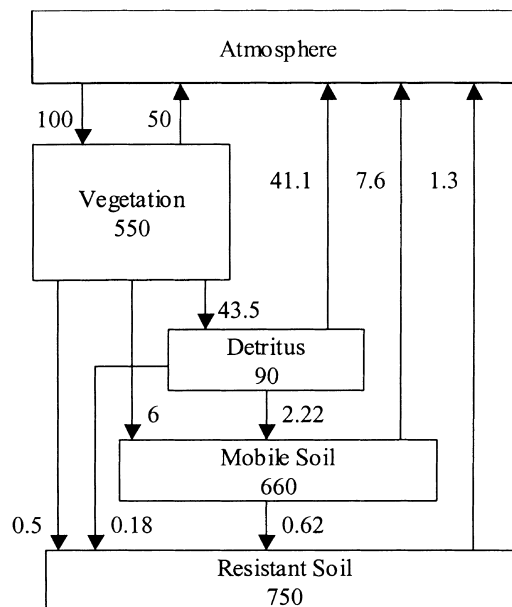


Fig. 5. Three-box soil model pre-industrial steady state. Reservoirs are in gigatons of carbon, GtC (10^{15} g), fluxes are in GtC yr^{-1} . The atmosphere and ocean are as in Fig. 1. The 3-box soil scheme is adapted from model 8 of Harvey (1989). The initial soil reservoir sizes were adjusted by a constant scaling factor to retain a total initial soil reservoir of 1500 GtC and fluxes were scaled to maintain total initial transfers of 50 GtC yr^{-1} from vegetation to soil and from soil to atmosphere. The resulting residence times are 1.8 yr for detritus, 80 yr for mobile soil and 577 yr for resistant soil.

to have the same temperature dependence (Lloyd and Taylor, 1994) and to be linearly dependent on the donor reservoir size, as in eq. (6). This isolates the effect of partitioning the soil reservoir, which strengthens the current land carbon sink such that the CO_2 fertilisation effect had to be weakened ($K_M = 96$ ppmv).

Fig. 6 shows the future response (dotted line) contrasted with the original model (bold line). Peak NEP is reduced, it becomes negative slightly earlier and a slightly lower minimum is reached, causing a ~ 20 ppmv increase in peak CO_2 (Table 8). The total soil reservoir then continues to shrink to $\sim 85\%$ of its original size after emissions have ceased, whereas it regains carbon in the original model. Hence the land surface remains a weak source of carbon dioxide (negative

NEP), increasing residual carbon dioxide and temperature levels. Warming drives slow shrinkage of the resistant soil reservoir over the ~ 600 year timescale determined by its residence time, which is counteracted by growth of the mobile soil reservoir after 2240. (Changes in the detrital carbon reservoir are relatively small but follow the same pattern as mobile soil.)

In a second experiment, respiration from the resistant soil reservoir was assumed to be independent of temperature, thus removing any tendency for this reservoir to shrink. In this case, $K_M = 89$ ppmv. The dashed line in Fig. 6 shows the result. The maximum land carbon source is reduced and predicted peak CO_2 is little altered from the original model (Table 8). The resistant soil reservoir remains close to a constant size, but

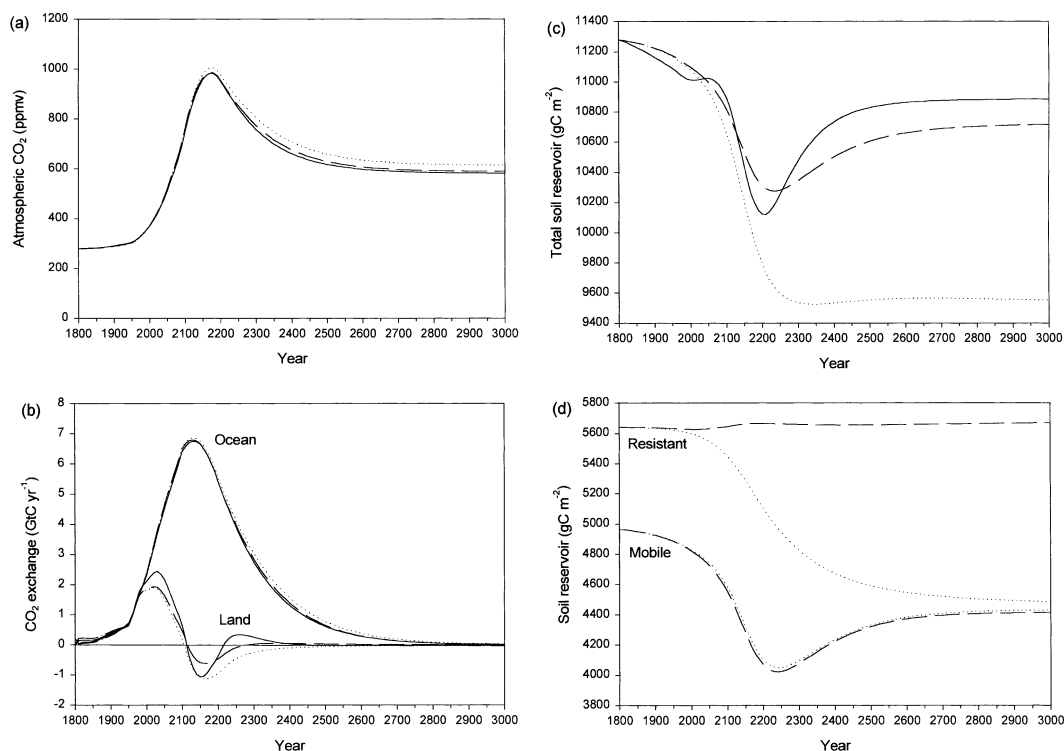


Fig. 6. Soil sensitivity experiments. The effect of soil structure and response to warming is explored for the extended "business as usual" emissions scenario 1. The solid line is the original model without ocean surface warming (version (iii), $K_M = 120$ ppmv), which provides a control. The dotted line is a model with the soil subdivided into three reservoirs (Fig. 5), each with the same response of respiration to temperature. The dashed line is the three-box soil model with respiration from the resistant soil reservoir assumed to be independent of temperature. (a) Atmospheric CO_2 . (b) Ocean-atmosphere and land-atmosphere net CO_2 exchange. (c) Total soil reservoir. (d) Resistant and mobile soil reservoirs for the versions of the model with the soil subdivided.

NEP still recovers more slowly than in the original model because of the longer timescale of response of the mobile soil reservoir, and the final size of the total soil reservoir is reduced. The 3-box soil system is worse at storing carbon than the original 1-box system because $\sim 80\%$ of the litter flux from vegetation passes very rapidly through the detritus reservoir and is returned to the atmosphere. Only $\sim 20\%$ of the litter flux has the capacity to be stored for a significant length of time, so it takes longer for the litter flux to replenish the majority of the soil.

Experiments with a 3-box vegetation scheme, adapted from model 8 of Harvey (1989), which distinguishes ground vegetation, non-woody tree parts and woody tree parts, caused counteracting increases in the magnitude of both maximum and minimum NEP and hence little effect on peak CO_2 (<10 ppmv increase, results not shown).

5.3. *Burning different amounts of fossil fuel in the same time period*

The sensitivity of NEP to the trends in future CO_2 emissions is now explored for the original model without warming of the ocean surface (this choice makes little difference to the predicted size of the land carbon source). First, the effect of burning different amounts of fossil carbon in the same time period is considered. For 1990–2100, in addition to the “business-as-usual” scenario IS92a for total emissions (discussed above), a “minimum emissions” scenario IS92c and maximum emissions’ scenario IS92e are considered (Table 4). These cover the possible extremes (c and e) and an intermediate path (a). Future land-use emissions are assumed to be the same in each case (to remove any sensitivity to this). From 2100 to 2200 emissions are assumed to decline linearly to zero in each case giving long-term scenarios 1, 2 and 3 (Table 5). For the “optimistic” scenario 2, a linear decline in emissions from 2100 to 2200 has a very similar gradient to the rate of decline in emissions at the end of the 21st century. However, for the “pessimistic” scenario 3 (and “business as usual” scenario 1) it requires a radical about-turn in emissions trends at 2100.

Fig. 7 shows the results. Peak and residual levels of atmospheric CO_2 and temperature vary to first order with the total amount of carbon added to the atmosphere from fossil fuel burning, but peak

levels also depend on the behaviour of the terrestrial carbon sink which is sensitive to the forcing. Peak levels of CO_2 are ~ 500 ppmv in 2130 for scenario 2 and ~ 1600 ppmv in 2180 for scenario 3. Corresponding peak temperature increases are $+2.3$ K in 2140 for scenario 2 and $+8.2$ K in 2190 for scenario 3. In year 3000, for scenario 2 the residual level of CO_2 is ~ 380 ppmv (airborne fraction $\sim 18\%$) and temperature $+1.3$ K, for scenario 3 CO_2 is stabilising at ~ 870 ppmv (airborne fraction $\sim 29\%$) and temperature at $+4.9$ K.

Under the “pessimistic” scenario 3, land ecosystems initially become a slightly larger carbon sink, peaking at 2.9 GtC yr^{-1} in 2030, but after 2090 become a much larger carbon source, with a minimum at -3.8 GtC yr^{-1} in 2140. The transition from terrestrial carbon sink to carbon source occurs at a slightly higher level of temperature, $+4.6$ K and CO_2 (~ 930 ppmv) under “pessimistic” scenario 3 than “business as usual” scenario 1. The “switch” is more rapid for the more extreme forcing — although the sink starts to decline later it becomes a source earlier. Peak temperatures are so high that they begin to reduce NPP (i.e., temperature reduces NPP more than CO_2 increases it), and this adds to the tendency for terrestrial ecosystems to lose carbon. The total carbon sink (ocean sink plus NEP) collapses from a peak of 8.4 GtC yr^{-1} in 2070 to 5.4 GtC yr^{-1} in 2140, a value less than that under “business as usual” (scenario 1) forcing at this time. Whilst they are a carbon source, land ecosystems lose a total of 270 GtC , mainly from the soil reservoir. During the preceding interval (1800–2100), when land ecosystems are a carbon sink, they gain 336 GtC due to positive NEP but lose 221 GtC due to land-use change, a net change of $+115 \text{ GtC}$. Thus, the land carbon source is of similar magnitude to the preceding land carbon sink, and because of the additional carbon loss due to land-use change, terrestrial ecosystems shrink below their pre-industrial carbon storage during the period of negative NEP. NEP recovers to provide a weak carbon sink after 2220 (but it is much smaller than the ocean sink).

Under the “optimistic” forcing scenario 2, terrestrial ecosystems never become a large source of carbon. Soil respiration does exceed NPP slightly after 2140, providing a weak carbon source (of greatest strength 0.4 GtC yr^{-1}), but shrinkage

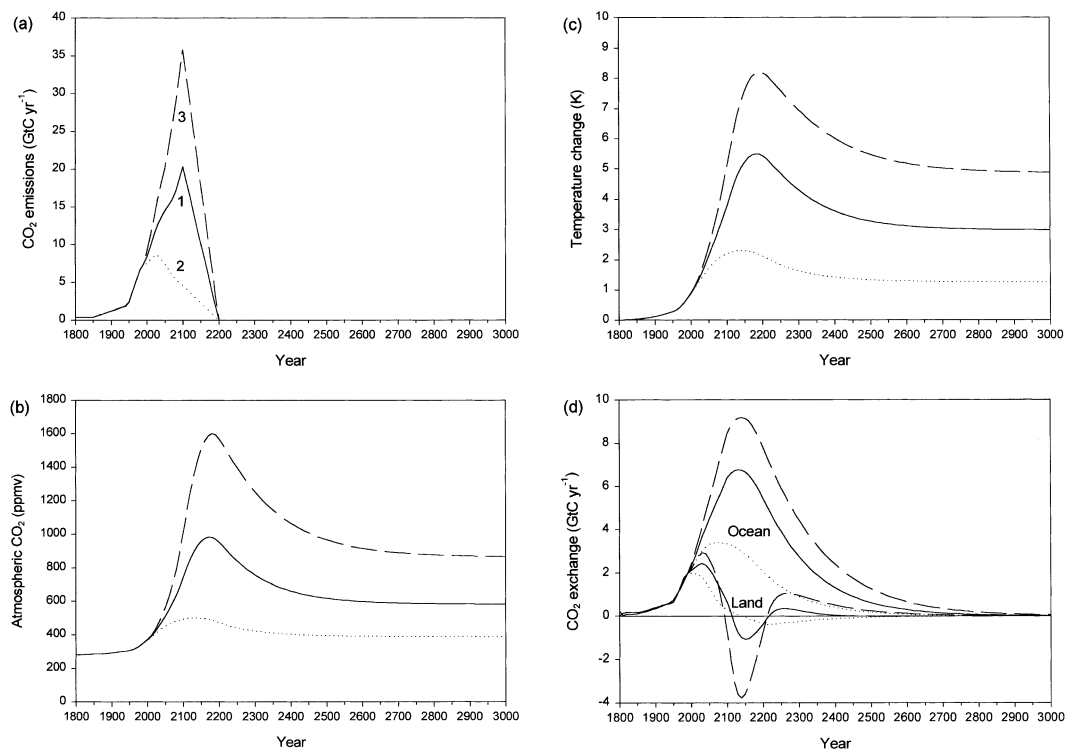


Fig. 7. Effect of burning different amounts of fossil fuel in the same time period. The results of forcing the model without ocean surface warming ($K_M = 120$ ppmv) with three different forcing scenarios, in which CO₂ emissions cease in 2200: (a) Total CO₂ emissions scenarios 1 (solid line), 2 (dotted line) and 3 (dashed line) (detailed in Table 5). (b) Atmospheric CO₂. (c) Temperature change. (d) Ocean-atmosphere and land-atmosphere net CO₂ exchange.

of the soil carbon reservoir is minimal compared to the other forcing scenarios. Warming alone is never great enough to drive soil respiration above NPP, although it does tend to reduce NEP. A carbon source is generated principally because CO₂ and NPP are declining and hence terrestrial carbon stores must adjust toward lower steady-state sizes.

The amplifications of peak CO₂ and temperature due to the effects of warming on terrestrial ecosystem processes have been quantified using the method in Subsection 5.1.2. (Table 6). Under “optimistic” scenario 2, peak CO₂ is increased by ~3% (16 ppmv) and peak temperature by ~7% (0.15 K). Under “pessimistic” scenario 3, peak CO₂ is increased by ~16% (225 ppmv) and peak temperature by ~12% (0.87 K). Comparing these results with those for scenario 1, under more extreme forcing, greater warming generates a greater amplification of peak CO₂ and temper-

ature. However, the amplification of warming is suppressed relative to the amplification of CO₂, because the climate sensitivity to changes in CO₂ is reduced at higher CO₂ levels.

5.4. Burning all fossil fuel in different time periods

In constructing future emissions scenarios it is important to consider the amount of fossil fuel available for economically viable extraction. The total recoverable fossil fuel reserves are estimated at 3400–7500 GtC (Sundquist, 1985) with 4000 GtC being a popular “ball park figure” (Sundquist, 1993). If IS92e is followed, ~2300 GtC of “fossil” carbon will have been burned by 2100 at which time emissions will be ~36 GtC yr⁻¹. For a total reserve of 4000 GtC, this rate of burning could only persist for 47 years beyond 2100 before the reserve was exhausted, and it therefore seems unlikely that emissions

would persist at such a high rate. The extension of IS92e presented above (scenario 3) involves a decline in emissions after 2100 and the burning of just over 4000 GtC. These considerations led to the generation of scenarios based on the assumption that all existing fossil fuel reserves will be burned, but exploring the effect of burning the same amount of fossil carbon over different time periods. (It seems unfortunately plausible that all fossil fuel reserves that are economically viable to extract will eventually be used up, if we remain dependent on them for the majority of our energy supply.) Total fossil fuel reserves are initially assumed to have been 4000 GtC (Sundquist, 1993). For this inter-comparison three scenarios for 1990–2100 are adopted: IS92a, constant emissions at 1990 levels and IS92e (IS92c suggests that not all the reserve would be burned). For the period beyond 2100, a linear decline in emissions is

assumed in all three cases, ending in 2332, 2926 and 2194 respectively, giving emissions scenarios 4, 5 and 6 (Table 5). Constant emissions at 2100 levels followed by “switching off” emissions have also been explored, but the discontinuity of abruptly reducing emissions to zero seems less realistic.

The results for burning 4000 GtC of fossil fuel reserves in different ways are shown in Fig. 8. The manner in which fossil fuel reserves are burned makes a significant difference in peak levels of CO₂ and temperature, although the eventual (residual) level of CO₂ (840 ppmv, airborne fraction ~28%) and temperature (+4.7 K) is the same. As might be expected, the highest CO₂ and temperature levels are achieved for the most rapid burning of a given amount of fossil fuel (scenario 6). The increase in peak CO₂ due to burning over the shorter time period (scenario 6), relative

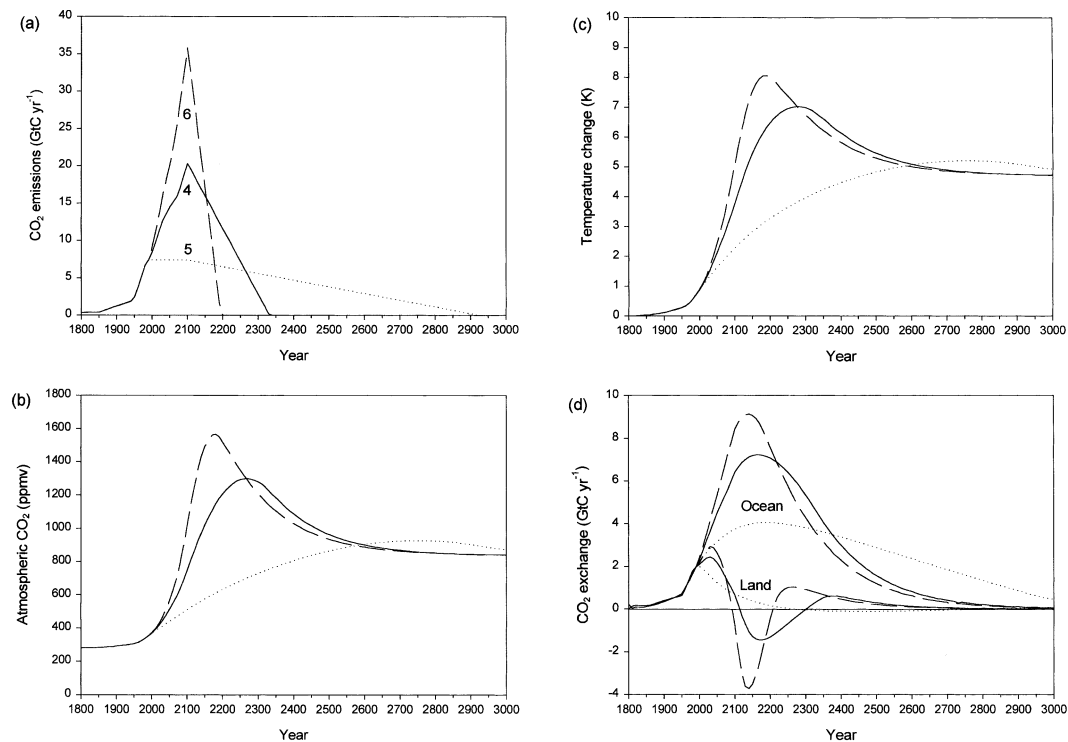


Fig. 8. Effect of burning the same amount of fossil fuel in different time periods. The results of forcing the model without ocean surface warming ($K_M = 120$ ppmv) with three different forcing scenarios, which involve burning a total fossil fuel reserve of 4000 GtC. (a) Total CO₂ emissions scenarios 4 (solid line), 5 (dotted line) and 6 (dashed line) (detailed in Table 5). (b) Atmospheric CO₂. (c) Temperature change. (d) Ocean–atmosphere and land–atmosphere net CO₂ exchange.

to the longer time period (scenario 5) is ~ 640 ppmv (from ~ 930 ppmv to ~ 1570 ppmv) and the increase in peak temperature is ~ 2.9 K (from ~ 5.2 K to ~ 8.1 K). In all scenarios, the peak levels are amplified significantly by the effect of warming on terrestrial carbon storage (Table 6, results from the method in Subsection 5.1.2). Amplification of peak CO_2 is greatest ($\sim 16\%$) for the most rapid forcing (scenario 6), but the fractional amplification of peak temperature is similar ($\sim 12\%$) for scenarios 4 and 6. A significant terrestrial carbon source is generated in both cases and is greatest for the most rapid forcing (scenario 6). Under the most gradual burning of the fossil fuel reserves (scenario 5) a slight land carbon source is generated when warming reaches 3.8 K at ~ 700 ppmv CO_2 but it is only of order -0.1 GtC yr^{-1} . This is despite the fact that temperatures reach a peak of $+5.2$ K at ~ 930 ppmv CO_2 (in 2750), which is sufficient to trigger a significant land carbon source under more rapid forcing. The reason is that when CO_2 and temperature are changing gradually, vegetation and soil carbon stores can track more closely their equilibrium state for a given forcing, which by definition has zero net ecosystem productivity (under scenario 5, NEP is close to zero from 2200 onwards). The lack of a significant carbon source helps reduce the amplification of global warming to $\sim 8\%$.

These results and those of the previous section reveal that the magnitude of maximum and minimum NEP is dependent on the rate of forcing. Furthermore, the threshold of CO_2 and temperature at which NEP switches from being positive to negative increases with the rate of forcing.

6. Conclusions

Natural land and ocean carbon sinks are currently buffering the rise in atmospheric CO_2 that is being driven by anthropogenic CO_2 emissions. This enhanced carbon uptake is largely caused by the increase in atmospheric CO_2 , providing a direct negative feedback, which also tends to reduce the degree of global warming. However, the model developed here suggests that global warming could have opposing effects, tending to generate a carbon source on land and a reduced carbon sink in the ocean. These responses tend to amplify the rise of CO_2 and increase temperature,

providing an indirect positive feedback. Although the positive effect of increasing CO_2 on net ecosystem productivity currently dominates over the negative effect of increasing temperature, if future global warming is relatively rapid and exceeds ~ 4.5 K, the model predicts a transient carbon source from warming-induced increases in soil respiration overtaking CO_2 induced fertilisation of plant growth.

If forest re-growth or nitrogen fertilisation contribute significantly to the current land carbon sink, this implies a weaker CO_2 fertilisation effect and a greater future carbon source. When the long residence time of some soil carbon is included in the model, the carbon source is predicted to persist for longer. If the temperature response of photosynthesis is stronger or those of plant or soil respiration are weaker than assumed in the basic model, then the predicted carbon source is reduced. However, the converse is also true and the assumed temperature responses lie in the middle of the range of estimates.

The modelled terrestrial carbon source is greatest for the most rapid consumption of the largest amount of fossil fuel reserves. The positive feedback on CO_2 and temperature due to CO_2 release from terrestrial ecosystems is a modest amplifier of global change, potentially increasing peak levels of CO_2 by up to $\sim 16\%$ and peak levels of temperature by up to $\sim 12\%$. The land carbon source can be minimised if global warming remains below ~ 4.5 K or if it occurs very gradually (in which case warming can exceed ~ 4.5 K without generating a significant land carbon source). Limitation of peak warming below ~ 4.5 K can be achieved by following emissions scenario IS92c. Alternatively, a gradual rate of global warming can be achieved by burning the available fossil fuel reserve very slowly (for example, by holding emissions at 1990 levels until 2100 and then reducing them linearly thereafter). However, "business as usual" (IS92a) to 2100 followed by a linear decline in emissions to zero at 2200 is sufficient to generate a significant land carbon source. The ocean carbon sink appears to be more robust than the land carbon sink and may be more important in the long term. However, warming-induced reduction in the solubility of CO_2 can be expected to suppress the ocean carbon sink and act as a further amplifier of increases in atmospheric CO_2 and temperature.

The simulations presented should not be taken as comprehensive predictions of global change. Many important anthropogenic forcings are not considered, including methane emissions, sulphur gas emissions and changes in land-surface albedo due to land-use change. A number of mechanisms that may affect land carbon exchange have not been included (Braswell et al., 1997), for example, warming increasing nutrient availability (Schimel et al., 1994) or delays in the migration of vegetation (Smith and Shugart, 1993). Furthermore, other potentially important climate feedbacks are not modelled, including those involving changes in sea ice cover, land-surface albedo or dimethylsulphide emissions.

The model assumes that terrestrial carbon cycling responds to global average surface temperature when in reality it responds to local temperature and moisture conditions (Saleska et al., 1999). Inclusion of these effects could alter whether a land carbon source is predicted, its timing, its magnitude and at what levels of CO₂ and temperature the critical threshold is reached. If processes other than rising CO₂ act to increase temperature then the critical threshold may be reached at a lower temperature (because there is a lesser CO₂ fertilisation effect on growth to be counteracted). Equally, if CO₂ is increasing but other processes tend to damp the rise in temper-

ature, the threshold would be reached at a higher level of CO₂.

The broader implication of this study is that when the Earth is relatively cool (below ~20°C on average), land ecosystems provide a short-term negative feedback on changes in atmospheric CO₂. However, if other processes drive atmospheric CO₂ and global temperatures above critical threshold levels then land ecosystems can provide a short-term positive feedback on changes in CO₂ and temperature. During the last period of rapid warming of the Earth, the transition from the last glacial maximum to the Holocene (~18–10 kyr BP), terrestrial ecosystems remained in the negative feedback regime, tending to damp the increase of CO₂ and accumulate carbon on land (Bird et al., 1994). However, in earlier epochs, perhaps the beginnings of the Carboniferous or the Cretaceous (Berner, 1990), CO₂ and temperature may have risen rapidly enough and to high enough levels to trigger a transient positive feedback.

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